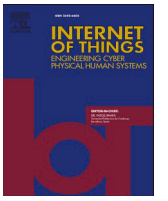
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A Comprehensive IoT edge based smart irrigation system for tomato cultivation

Rohit Kumar Kasera *, Tapodhir Acharjee

Department of Computer Science & Engineering, Triguna Sen School of Technology, Assam University, Silchar, Assam, 788011, India

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ABSTRACT

Agriculture industry is the primary engine for a country's economic development. Growing crops using minimum irrigation water is a major challenge for farmers. In conventional farming, crops may be affected by various diseases due to inadequate irrigation scheduling. Recent proposals have suggested using Edge-IoT, AI, and distributed computing to accelerate the inference procedure utilized in smart irrigation applications. The use of resource-constrained edge servers and edge devices used to deliver smart agriculture applications can cause latency-sensitive workloads to interfere with one another. To address this issue, we design a long-range (LoRa) edge IoT computing-based sustainable and customized smart irrigation framework to capture the real-time data of tomato plants. This helps in automatic underground drip irrigation scheduling. This also predicts total water demand and usage, and measure plant growth status. The edge-IoT cloud data transmission control and optimization has been enforced using Smart irrigation data optimization and robust transmission (SIDORT) Message Queuing Telemetry Transport (MQTT) system. We develop a hybrid algorithm named Linked least traversal (LLT) for machine-to-machine communication (M2M). Also, a Reinforcement learning (RL) based Optimal Soil Wetness Closeness Policy (OSWCP) for irrigation scheduling has been proposed. The performance of the proposed smart irrigation models has been validated through extensive experiments using real-time data in which OSWCP performance has been measured at a 97.88 % accuracy rate. Additionally, a comparison of our proposed architecture has been accomplished by resolving the existing smart irrigation system challenges.

1. Introduction

The agriculture sector is continuously becoming smarter by adopting cutting-edge technologies to increase production and meet global demand. One such technology is Wireless Sensor Networks (WSN), which allows smart devices to manage and monitor crop development in real-time. Agriculture practices today use smart technologies that are affordable, accessible, and have minimum negative environmental effects. To increase production, farmers need to use smart agriculture technologies in various agriculture operations of pre-, during, and post-harvesting stages [1,2]. Most of the water is used in irrigating the soil and roots of the crops during the various stages of crop's growth. Tomato is one of the very heavily used vegetables across India and globally. Balanced irrigation is needed at each developmental stage of tomato crops [3]. According to the Food and Agriculture Organization (FAO) report from 2021 – 2022 tomato production was 20,694,000 tones in India [4]. 91 % of India's total tomato production comes from its thirteen central

* Corresponding author.
E-mail address: rohitkumar.kasera@aus.ac.in (R.K. Kasera).

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Table 1
Optimal Soil moisture, temperature, and humidity in each growth stage of tomato plant.

Stage	Temperature [°C]	Humidity [%]	Soil moisture [%]
Seed Germination	Minimum: 11, Maximum: 34, Optimal: 16–29	60–75	20–50
Seedling growth	Minimum: 18, Maximum: 32, Optimal: 21–24	60–85	50–60
Development stage	Minimum: 18, Maximum: 32, Optimal: 21–28	70–85	60–85
Fruit set	Minimum: 18, Maximum: 30, Optimal: 20–24	75–85	65–85
Red color germination	Minimum: 18, Maximum: 30, Optimal: 20–24	75–85	60–80

producing states. Production was predicted to have decreased 6.4 % in 2017–2018 compared to the previous year, and it was estimated to have decreased to 5.9 % in 2021–2023 [5]. Tomato crops are susceptible to a range of environmental factors during their development phases. These include unbalanced irrigation system, flooding, excessive saltwater, drought, below-normal temperatures and insufficient nutrient level of nitrogen (N), phosphorous (P) and potassium (K). Flower end root, skin bursting, untimely blight, stem curling, sunburn, etc. are few of the frequent consequences [6]. Maximum tomato yields is possible through the appropriate irrigation methods that maintain soil moisture levels, humidity, and temperature [7,8].

1.1. Challenges of IoT enabled irrigation

Manual watering is the most common method for irrigating crops. But this method results in too much water wastage, and it is challenging to monitor and distribute water evenly throughout every corner of the field [9]. Even the current automated irrigation system is not capable of preserving wastage of irrigation water. According to the existing research the optimal range of soil nutrient value for the tomato crop is between 40 and 120 for nitrogen (N), 30–90 for phosphorus (P) and 80–130 for potassium (K). Based on existing research on the automated irrigation system [10–12], Table 1 shows optimal ranges of temperature, humidity, and soil moisture for each growth stage of tomato plant. If these values are below or above the threshold, the tomato plants and fruits can be affected by diseases and by analysing this ranges, the proposed irrigation scheduling algorithm have been designed by field capacity (H) and soil moisture level (B).

Internet of Things (IoT) can be used to develop an automated drip irrigation management system for tomato crops that will reduce water wastage and easily track soil moisture levels across a farming field. In today's environment, reducing water wastage in agriculture and implementing drip intelligent irrigation systems are essential for raising agricultural output [13]. Various IoT based irrigation scheduling [14] algorithms have been designed using fuzzy logic [15], AI based intelligent irrigation system [16], "Crop water stress index" (CWSI) [17], "Evapotranspiration" (ET_C or EVT_C) [18] based water requirement, solar energy power based irrigation system [19], "ZigBee" [20] etc. But these existing methods still have some shortcomings in terms of adaptability, optimal irrigation performance, computing speed, storage etc. It is extremely challenging to develop an IoT based irrigation system using approaches, such as EVT_C, or CWSI, to determine the volume of water required for a given crop at a given stage of growth across the field based on soil moisture and specific climatic circumstances. The accuracy of the CWSI approach is susceptible to a range of climatic parameters, including soil moisture, wind speed, humidity, and sun radiation. If soil moisture isn't consistent, it can have major consequences on the accuracy of stress measurements due to continuous change [21]. EVT_C is the total amount of moisture released through transpiration from a plant via pores and also evaporated from the soil surface level. The tomato plant water and nutrient demands can be balanced using soil moisture and the EVT_C method. It can be used for measuring various parameters such as Temperature, Humidity, Wind speed, Solar radiation, Elevation, Latitude, Longitude, rainfall and nitrogen (N), phosphorus (P), potassium (K) for specific irrigation management. It is easily calibrated based on various stages of crop growth, providing accurate and scientifically rigorous estimates of crop water requirements [22], and N, P, K level.

1.2. Motivation and contribution

The design and deployment of a smart irrigation system via traditional IoT-based system and using multiple types of technologies is a viable approach to make farming easier and more sustainable. But the aforementioned systems operate on a fixed network, need more computing power, storage, and cost. This results in improper management of irrigation scheduling time because of continuously changing environmental factors, soil moisture variations, significant delays in data collection, and bandwidth restrictions. These issue occurs in large remote areas with limited connectivity to the global internet. There isn't enough database storage and processing power on a single controller device to transfer data to a cloud server. Continuously growing data leads to overflow of data. As a result, analysing and cleaning valuable data takes a lot of time, and effort.

Monitoring and controlling large farming fields' irrigation needs and determining which sensor data is lost or received from a single edge sensor node can be challenging with an IoT-based system. However, these issues still need to be fixed in the existing method to make automated irrigation system more efficient, robust and sustainable for low-income farmers. Each agricultural field can have an edge IoT gateway established as a decentralized edge network to overcome the aforementioned issue. The Arduino microcontroller operates as an edge sensor node with limited processing capabilities, interacting directly with several physical sensors to read data in an instantaneous environment. It can have n number of edge sensor devices as [Arduino 1, Arduino 2, ..., Arduino n]. This edge device will continuously read and transmit sensor data to its nearest edge IoT gateway as [Edge 1, Edge 2, ..., Edge m]. The Edge servers are equipped with high computing power, storage facilities, and internet connectivity. It is capable to hold all farming data in

Table 2
Summary of existing IoT-based irrigation algorithms and their flaws concerning the different technologies standard.

Reference	Objectives & contributions	Algorithm	Hardware & connectivity used	Results	Research gaps
[29]	Sensors are used to gather data on soil moisture, air temperature, humidity, water flow, and light intensity. Calculate the amount of water that tomato crops require and to publish the data to a cloud server.	Deep learning consisting of long, short-term memory (LSTM) and Gated recurrent units (GRU)	NRF24LO1 module, WIFI module with HTTP server	The platform could boost irrigation efficiency by up to 30%	Lack of features in terms of the type of irrigation, total water requirement for each growth stage, plant growth monitoring, adaption for large farming fields, long-range data transmission, and stable and robust transmission from edge server to cloud server.
[30]	A sophisticated IoT-based drip irrigation surveillance system for mustard leaf farming has been put into effect with dependence data and experiment.	The autoregressive (ARX), ARMX and Box Jenkins (BJ) standards based on the lowest possible order, consistency, and estimation of the volumetric water concentration of the soil have been performed.	Raspberry pi with WIFI internet	Based on the ARMX model, the accuracy is 91.31%	No automated system to manage the water storage in tank. Lack of optimizing irrigation scheduling features based on soil, field, water pressure, and crop type for estimating water volume requirements. The system is limited to small farming fields. Features of reliable, low-power IoT connectivity between the physical and cloud layers are lacking in the proposed architecture
[31]	Constructing an affordable sophisticated watering calendar that considers both the soil moisture content and indicator of agricultural water stress.	Fuzzy logic-based irrigation	NRF24L01with Arduino DUE controller	Significant increase in irrigation precision and trustworthiness, which boosted the soil and plant water status.	Lacking a suitable strategy for establishing water measurements. Several factors should be analysed, including crop type, irrigation time, field size, and water consumption. A communication network protocol is not used to enable long-distance and stable data transmission from the sensor layer to the cloud.
[32]	An IoT-driven autonomous watering and farm tracking operation is employed. Several modules the unitary detector pole (USP), water distribution section (IU), and sensing unit (SIU) have been built for managing the watering system.	Neural network (NN)	ESP8266 with MQTT and HTTP server	Through MQTT and HTTP servers, SIU is used for maintaining sensor data for remote data observation. This practice of irrigation is shown to save 67% of water overall.	Irrigation method that is employed is not covered. Other crops type still need to be analyzed to see how well the water timetable can be managed. The lack of a soil moisture and climate-based analysis of over- and under-irrigation.
[33]	Watering system is proposed that regulates irrigation using Arduino.	Soil moisture sensor-based irrigation	Arduino Uno with serial communication	The drainage valves are commandeered by a microprocessor, and sensors are used for gaging the water content and temperature underlying the soil.	The system has not been extensively tested based on crop type. Validation hasn't been conducted with alternative irrigation methods, and irrigation scheduling is not automated or optimized. The sensors used seem inadequate.
[34]	A free, open-source Internet of Things platform is used to oversee smart irrigation systems.	The algorithm is administered taking advantage of edge computing and/or cloud computing infrastructure	Arduino Uno, RS-485 module, TCP protocol with 4 G mobile internet	The structure integrates an irrigation simulation that estimates wet bulb size and soil water regulation to decide on the optimum drip gardening proximity.	On the platform's long-term performance, the system offers no information. Additionally, it makes no mention of the possibility of applying the

(*continued on next page*)

Table 2 (*continued*)

Reference	Objectives & contributions	Algorithm	Hardware & connectivity used	Results	Research gaps
		and it relies on the FIWARE architecture.			platform to other agricultural crop environments, such as certain types of fruits and vegetables. Lack of implementation of which type of network and data transmission method applied.
[35]	An internal wireless measurement network that is efficient has been created using an adaptable watering system.	A low-cost agriculture sensor is designed to measure soil water level	Zigbee, GPRS	Compared with the traditional irrigation, it improves crop productivity by 10.21%.	It is yet unclear how well the system would function in a large-scale industrial scenario because it has only been tested in a small-scale field trial. For farmers with limited resources, the system requires a steady internet connection, which may not be available in all remote locations. No intelligent method is used for scheduling irrigation.
[36]	An affordable sensing soil moisture and a wireless connection powered by LoRaWAN are utilized by the system to track and handle sub surface irrigation for tomato plant.	Soil moisture-based irrigation.	LoRaWAN with Arduino uno	Compared to conventional irrigation systems, the results of the method were able to greatly improve crop yield.	Installing and maintaining the system may be costly. Lack of IoT-based agronomical metric analysis for plant biomass advancement, flowering, fruit counting systems, total volume of water requirement, soil health, and crop growth. Limited use of AI method for irrigation scheduling. Absence of stable, strong, and distributed data transmission between the sensor layer and the cloud layer.
[37]	For sweet corn crops, comparisons have been made between IoT based soil moisture and evapotranspiration (ET_C) based technique evaluation.	IoT drip irrigation using evapotranspiration	ESP32 WROOM-32 WIFI internet, HTTP server	In terms of increasing the yield of crop, ET_C based irrigation has outperformed.	Lack of estimating total water requirement, scheduling irrigation, soil nutrient level at each growth stage of the crop, and real-time data transmission from the physical layer to the cloud layer. Only suitable for small fields of agriculture.
[38]	For farmers, a sustainable irrigation approach has been devised. Wherein real-time data are used to identify the optimal irrigation process.	Random forest, decision tree, and neural network	ESP32 with RFM95W module for data transmission is used through the MQTT protocol.	An accuracy of 84.6%, Random Forest (RF) accomplishes above the best ML models	A lack of IoT-based irrigation scheduling and real time data transmission between edge to cloud server, inadequate testing based on crop type, and inability to control overwatering and underwatering.

local database storage, perform adaptive computation methods for irrigation scheduling, make automatic irrigation decision and synchronously transmit it to the cloud server through a network communication protocol [23]. The entire Low-power wide area networks (LPWANs) transmission of edge servers is performed using a modified Edge-based "Message Queuing Telemetry Transport" (MQTT) [24] based on a distributed evolutionary computation framework (DEAP) [25,26] named as Smart irrigation data optimization and robust transmission (SIDORT). It is lightweight and has a low-power and robust transmission method. On edge and cloud servers, all data traffic is monitored and administered, including packets being sent, received, lost, delayed, and acknowledged.

This paper proposes an automated method for scheduling underground drip irrigation to increase tomato production. It monitors

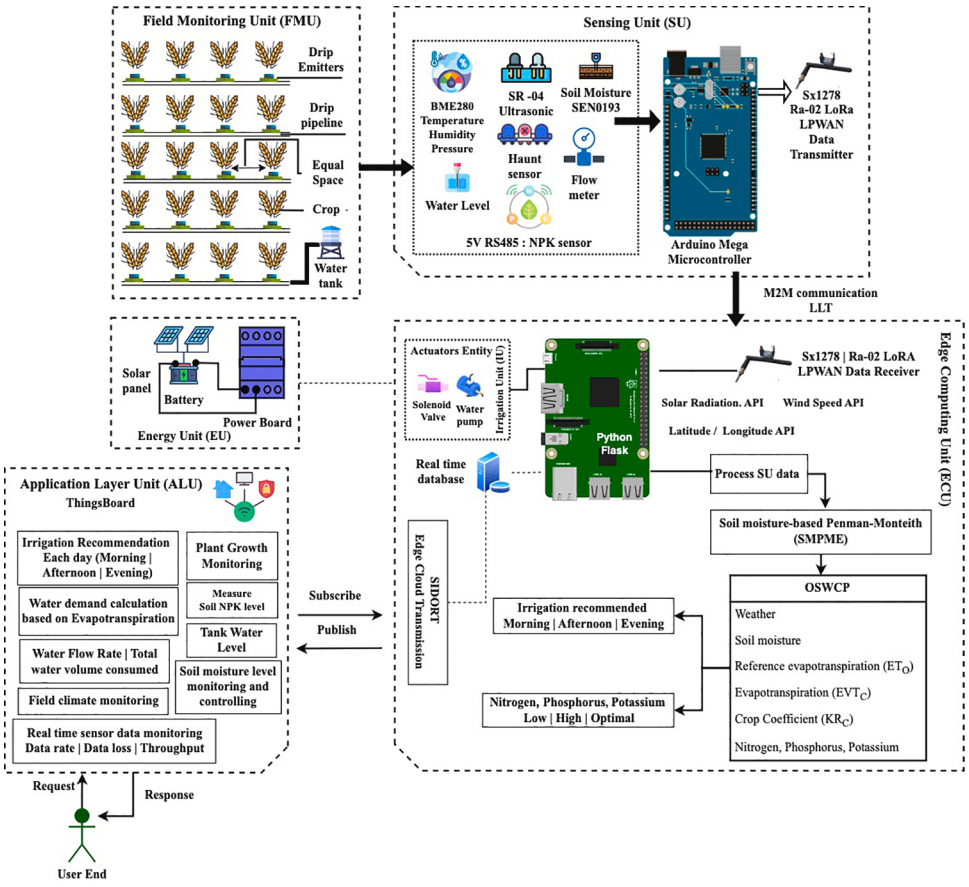


Fig. 1. Edge based architecture for irrigation and field monitoring system for Tomato cultivation.

tomato fields by estimating weather conditions, soil health, and total water demands. Furthermore, it computes day-to-day irrigation water consumption and monitors plant growth. The following list are the primary contributions made in this paper to increase the sustainability and efficiency of irrigation system:

1. This manuscript proposes a hybrid algorithm known as a Linked Least Traversal (LLT), which integrates the principles of a "Link budget calculator" (LBC) and a "Minimum spanning tree" (MST) to create an distributed long range wide area network (LoRaWAN) in the subSection 3.1. This algorithm is used to establish stable machine-to-machine (M2M) communication remotely between sensor nodes and an edge servers for transmitting sensor data from farming field (FF) to long distance.
2. It presents a design of a hybrid algorithm named as Optimal Soil Wetness Closeness Policy (OSWCP), which uses the "Penman-Monteith" and "Reinforcement Learning - Proximal Policy Optimization" (RL-PPO) methods for a distributed edge IoT-based adaptive underground drip irrigation scheduling system for tomato crops in the subSection 3.2. It recommends irrigation optimal during specified periods, optimizes the allocation of total water resources, measures and optimizes soil health and nutrient levels, and estimates tomato field climate conditions. It also states the implementation of a sub-system to observe weather condition, soil health, nutrient level, plant growth that will be used to gage a tomato plant's height at each stage from root to tip and build a water-level measuring module for farming fields (FF) water tank.
3. Lastly in subSection 3.3, this paper furnishes a lightweight communication framework to optimize and supervise accurate and knowledgeable data transmission between a distributive edge IoT and a cloud server. It uses the Smart irrigation data optimization and robust transmission (SIDORT) method through which data transmission is optimized with lower energy consumption and without data loss and delay.

The rest of the paper is organized in sections as follows: The existing smart irrigation systems algorithm, along with the benefits and drawbacks, is covered in Section 2. Proposed edge IoT architecture for developing customized smart irrigation system is discussed in Section 3. The overall results and analysis of the proposed automated customized irrigation algorithms experiment are presented in Section 4, along with a comparison with existing smart irrigation systems. In Section 5, the results of the proposed smart irrigation system are summarized with recommendations for further development.

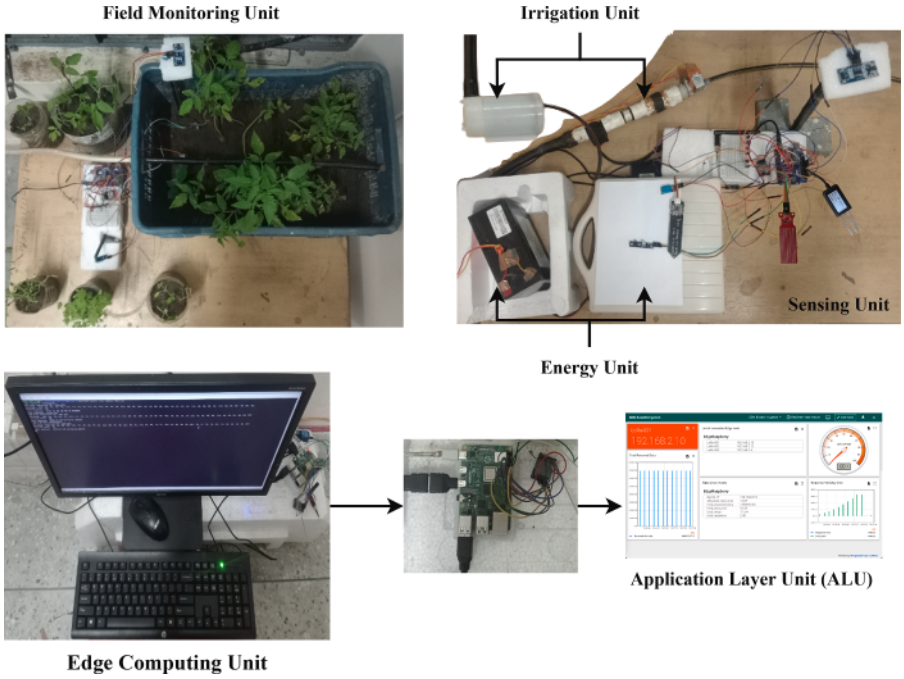


Fig. 2. Prototype setup of Edge based customizable irrigation systems.

2. Related work

Numerous research have been undertaken for irrigating [27–28] specific crops based on organizing, evaluating, and refining the system for IoT based drip irrigation. The following section discusses some of the significant existing systems, along with their benefits and comparative research gap in the Table 2.

The comparation table as shown in Table 2 depicts that existing approach lacks adaptive and real-time system performance based on customized irrigations system.

- Research on allocating water resources according to crop type, soil moisture, soil nutrients, and climate conditions is still essential.
- Comparative analysis of IoT-based automated irrigation methods between manual, soil moisture, and evapotranspiration is lacking. Real-time monitoring of crop growth, soil health, and water usage at each growth stage is still needed to be implemented.
- IoT architecture lacks real-time, robust, and stable data transmission at each layer, based on distributed Low-power wide area network (LPWAN) machine to machine (M2M) communications for remote regions, optimization of real-time data transmission between edge and cloud servers, and comparative analysis of algorithm performance.

3. Methodology

Our Proposed smart drip irrigation architecture is depicted module by module in Fig. 1. Here we present a distributed edge computing framework with IoT based network for underground drip irrigation scheduling and field monitoring systems, thereby improving their efficiency and performance. A prototype field monitoring unit (FMU) was established by sowing tomato seeds at a uniform spacing to track the crop's growth status using a variety of sensors and actuators. It used a solenoid valve (SV), a mini water pump, a relay module, an ultrasonic SR-04, Soil moisture SEN0193, YFS2–01 water flow meter, Hunt Sensor, RS-485 N, P, K sensor, and water level. An Arduino mega microcontroller board interfaces with the sensors installed in the FMU. The sensor gathers the FMU data on temperature, humidity, soil moisture (SM), plant height, total water consumed, nitrogen (N), phosphorus (P), potassium (K), and the water tank level. These data then transmitted to the edge IoT gateway as Raspberry pi 4 using the LoRa Sx1278 Ra-02 RF 433 MHz radio frequency module. The edge server receives the sensor data. Subsequently, sensor data is evaluated, and the suggested Optimal Soil Wetness Closeness Policy (OSWCP) algorithm is used to manage irrigation decisions by activating SV each day between specific hours (6–8AM, 12–1PM, 4–6PM) to irrigate the tomato crops. Automatic SV activation is decided based on the deep Reinforcement Learning (RL) with reward awarded to the agent. This reward is calculated based on the current state and selection of the action space by the agent of the environment. For example, if the time interval is between 7 and 8AM on a day, and state space current parameter Evapotranspiration is 300 mm/day, the soil moisture percentage is 55 %, and the crop coefficient value is 0.73, then set of actions selects the total amount of water needed value of 255 mm/day and positive reward is awarded, which triggered to activate the SV for irrigating. SV actuator will deactivate if the soil moisture value exceed the threshold and the total amount of water required is less than zero. The RS-485 N, P, K sensor is used to estimate the soil nutrient level as Low, High and Optimal based on the current

climate conditions and soil health status. The total volume of water consumed for a day is estimated by the water flow meter during irrigation. The water level sensor data is used to measure the water level in the tank. The mini water pump is activated to fill the tank when the combined value of the water level sensor and the total water in tank value is less than a certain threshold, and it is deactivated otherwise. For measuring tomato plant height, the ultrasonic "SR-04″ and haunt sensors are utilised. The FF operation data are stored in a “MongoDB” database and are then sent to a cloud server via the modified Message Queuing Telemetry Transport (MQTT) broker named as Smart irrigation data optimization and robust transmission (SIDORT) for user visualization and analysis. The distributed edge gateway will regulate and oversee the data sent, received, and loss of all sensor nodes in between these transmissions using LPWAN wireless internet.

In the Fig. 2, the prototype system is shown as divided in to six units, namely the field monitoring unit (FMU), irrigation unit (IU), sensing unit (SU), edge computing unit (ECU), energy unit (EU), and application layer unit (ALU). The real-time monitoring of FMU includes automatic management of irrigation, plant growth status, climate, water tank level, etc. An SU serves as a tool in this FMU process. Sensors discussed above are placed in the FMU adopting the SU, and real-time data is gathered from them for performing out automatic irrigation activities. Based on SU data, the ECU regulates every SU computation and action. ECU pre-process and computes all IU, EU, and SU data using data processing method, while EU manages the system’s overall energy or power. The Linked least traversal (LLT) algorithm is used to manage a stable M2M connection between Multiple Arduinos and Raspberry Pis in a large farming field. The irrigation plan was designed using a hybrid adaptive algorithm Optimal Soil Wetness Closeness Policy (OSWCP) by integrating the soil moisture-based “Penman-Monteith” [39] (SMPME) method with “Reinforcement Learning” (RL) using the “Proximal Policy Optimization” (PPO) [40] model. The number of factors which have been taken into considerations for developing this real time application for “during harvest” activities, include latitude, longitude, temperature, humidity, solar radiation, windspeed, elevation, crop height, soil nutrients, water flow rate, total volume of water consumed in milliliter (M/L) and various crop coefficient values for tomato crops.

For data transport over different networks to the cloud server from where ALU accesses data from the farming field, the ECU manages the FF data, sent, received, and lost. The OSWCP model recommends the irrigation policy each day in morning, afternoon and evening time. Whenever there is alarm of irrigation time the automatic OSWCP system verifies the current status of the climate, soil health, tomato crop development stage and EVT_C (total water requirement). If the current status reward is positive, it activates SV for irrigating tomato crop and notifies about the soil Nitrogen (N), phosphorus (P), potassium (K) level each day. Otherwise it will be deactivated. The OSWCP model keeps learning automatically at each time step based on the rewards for the current weather condition, soil moisture, and nutrients. On the basis of real-time data, the overall system tracking is performed. Through the ALU, the user can access information about IUs, FFs as well as EUs. The power sources used for testing the overall system is 12v solar power charging battery.

3.1. Decentralized M2M communication

IoT based communication network development is crucial for the creation of an irrigation scheduling and plant growth monitoring system that is entirely automated. There has been research on WSN that have been developed for irrigation and related applications like [41]. In our proposed hybrid Linked least traversal (LLT) algorithm it provides reliable, robust, and flexible connectivity to a distributed long-range (LoRa) edge IoT network, ensuring optimal communication with low power consumption and strong signal strength for all devices. An Arduino setup is installed in the tomato farming field, and a Raspberry Pi (RPI) edge IoT gateway setup is installed in the indoor room. The goal is to establish a point-to-point (P2P) wireless communication in a real time at a fixed distance and communication between two devices is conducted through a single-hop medium. The proposed LoRa based LLT algorithm employs the “LoRa SX1278” data transmitter and receiver module to establish machine to machine (M2M) communication for building a decentralized low power wide area network (LPWAN) IoT based edge computing server, in contrast to the existing minimum spanning tree (MST) [42] technique, which uses GPS, “ATSDM21”, RFM95W, and SX1301 module for M2M communication [43]. In this case, the RPI nodes are equal to R_1, R_2,....., R_m, while the set of Arduino nodes are equal to A_1, A_2,…, A_n. Each node has a uniform maximum transmission distance (MTD). The "LoRa Sx1278″ radio frequency module has a range of 100 to 15,000 metres (m) at its full capacity for data transmission [44]. For Arduino A_i and Raspberry R_j, the node’s coordinates are $C(A_i)$ and $C(R_j)$, respectively. The value minZ is used to minimize the overall weighted distance which is calculated using Eq. (1).

$$\text{MinZ} = \Sigma_{i=1}^{n}\Sigma_{j=1}^{m} k_{ij} * Ed\big(C(A_i)C(R_j)\big) \tag{1}$$

The k_{ij} is the weight which defines the communication link between the node A_i and R_j. There will be a minimum of one immediate connection within any subset of nodes A_i and R_j considering the sum of all weights (k_{ij}) for nodes in the subset must be larger than or equal to 1 which indicates that MST is linked. When $n > m$ and $k_{ij} = 1\ \forall$ (i,j) n and m, each Arduino has a single RPI connection, and vice versa. MTD $k_{ij} = 0$, if total distance Ed(C(A_i),C(R_j)) > MTD $\forall$ (i,j) in n and m. Received signal strength indicator (RSSI) [45,46] is acquired from LoRa transceivers module which is measured using Eq. (2), and it used for calculating total distance (Ed) as shown in Eq. (3).

$$RSSI = CP + IBA + IRA - \big(20\log_{10}(Ed) + 20\log_{10}(freq) - 20\log_{10}\frac{4\pi}{c}\big) \tag{2}$$

Algorithm 1
Linked least traversal algorithm for IoT edge-based communication.

Input: Arduino = $A_1, A_2, \ldots, A_n$, RPI = $R_1, R_2, \ldots, R_m$, MTD = 1000, RSSI
Output: MST communicating each Arduino with nearest RPI within maximum transmission range.
MST = G (), visited Arduino (V_ARD) = [Ø], visited RPI (V_RPI) = [Ø], priority_queue (PRQ) = [Ø]
nearest RPI (N_RPI) = map (Arduino = RPI)
min_dist = map (Arduino = Ed)
/* Any arbitrary Arduino as starting point */
start_ARD = A_1.
V_ARD = start_ARD
/* calculates initial distances */
for each R_j = 1 to m **do**
* calculate (calc) the distance based on RSSI value from start_ARD to R_j */
  Ed = calc (start_ARD, R_j, RSSI)
  if Ed <= MTD **then**
    N_RPI [start_ARD] = R_j
    min_dist[start_ARD] = Ed
    PRQ.add ((start_ARD, R_j), Ed)
  end
end
/* Visit all Arduino till it is not visited. */
while V_ARD ≠ A_i to A_n **do**
/* Extract the edge which has minimum distance from PRQ. */
  (Arduino, R_j) PRQ.minimumEdge
/ * if Arduino is already visited */
  if Arduino in V_ARD **then**
    continue
/* Add the edge in to MST */
  MST.addEdge(Arduino, R_j, weight min_dist [Arduino])
/* Mark particular Arduino is visited */
  V_ARD.add (Arduino).
/* Mark particular RPI is visited */
  V_RPI.add (R_j)
  for each neighboring A_i of R_j
    Ed = calc (A_i, R_j)
    if Ed <= MTD & Ed < min_dist [A_i] **then**
      N_RPI[A_i] = R_j
      PRQ.add ((A_i, R_j), Ed)
    end
  end
end

$$Ed = 10^{\frac{\left(CP+IBA+IRA-RSSI+20\log_{10}\frac{4\pi}{c}-20\log_{10}freq\right)}{20}} \quad (3)$$

In Eq. (2), CP corresponds to communication power, IBA is the signal strength of broadcast antenna, IRA is the signal strength of reception antenna, Ed is the distance, freq means the frequency of the transmitter module, 4 π is the spread factor of the free space path loss measure [47], and c is the speed of light, which is 3×10^8 m/s. So to calculate Ed Eq. (2) is rearranged in the Eq. (3).

Algorithm 1 seeks to connect each Arduino device to its closest Raspberry Pi edge gateway. Arduino, Raspberry pi (RPI), MTD, and RSSI are declared as input variables. The point to point (P2P) connection between Arduino and RPI nodes is tracked in a priority queue (PRQ). For each RPI (R_j), the RSSI value was used to determine the initial distance (Ed) between Arduino (A_1) and R_j. The most recent

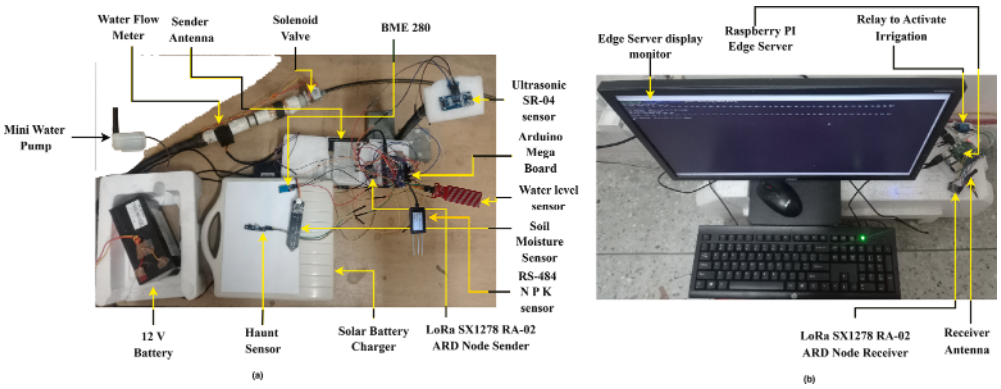


Fig. 3. Arduino node transmitter and receiver hardware model.

RPI (R_j) for Arduino (A_1) is amended with its maximum distance value if the calculated distance is less than or equal to the maximum transmission distance (MTD). As long as all Arduino nodes are not linked with RPI, the while loop will continue. In this process, the connection with the shortest distance is selected first from the PRQ. If an Arduino node is already connected, it moves on to the next connection, then a particular connection will link to the R_j if the Arduino node is not already hooked up to that particular R_j and it will be updated in the visited list of the Arduino (V_ARD) and RPI (V_RPI) list. For each adjacent A_i in the selected R_j, the Ed is measured. When the Ed is less than or equal to the MTD and the Ed is smaller than the current measured minimum distance (min_dist) of A_i, the connection is updated with the nearest RPI (N_RPI) for the neighboring Arduino A_i node. and is placed in a PRQ with the distance (Ed). Through these processes, all Arduino and RPIs will establish P2P connections. These will result in distributed communication and a reduction in transmission load.

A time complexity analysis of the proposed Linked least traversal (LLT) algorithm showed that the initialization of variables takes O $(n + m)$ time. In contrast, the starting distance between Arduino and RPI takes $O(m) + O(m \log m) = O(m \log m)$. The outer loop runs $O(n)$ times for connecting Arduino and RPIs, whereas the inner loop updates the priority queue for each m possible connection under the while loop up to $O(n^2 (\log m))$. So overall complexity can be estimated as $O(n + m) + O(m \log m) + O(n^2 \log m) = O(n^2 \log m)$ in the worst case. A prototype hardware setup for distributed edge IoT-based network for automatic irrigation and plant growth monitoring systems is depicted in Fig. 3.

The hardware module of the Arduino LoRa-based transmitter, shown in Fig. 3(a), is interfaced with an array of sensors to record data on tomato plant development and irrigation from FMU. This LoRa Arduino edge sensor node is utilized to connect with the RPI LoRa-based edge server within a specific range. Fig. 3(b) is a representation of a LoRa-based edge server. It receives Arduino node data for processing if any Arduino edge node is detected within its range. Each edge server's data are pre-processed, cleaned, and computed during the data processing stage, where RWD is the raw data collected from the Arduino node, and EPD_i is the processed data at the ith edge server (RPI).

$$EPD_i = f_i(RWD) \tag{4}$$

In Eq. (4), f_i is the processing function at the ith edge server (RPI), which is put to use by the edge server to carry out computing, cleaning, and pre-processing.

3.2. Irrigation scheduling and crop growth monitoring

Linked least traversal (LLT) algorithm is being used to create a decentralized edge-based computing server. Data from FMU sensor nodes are sent through SU to an ECU. To use the Optimal Soil Wetness Closeness Policy (OSWCP) method for scheduling the irrigation for the tomato crop, these data are pre-processed and put through calculation. The actuators SV automatically activate and irrigate the tomato crops based on the OSWCP prediction. In the IU, the pipeline has been installed with one drip line per row across the FMU [48], where an online dripper is installed by drilling a tiny hole measuring 0.8 mm s (mm) in diameter where a tomato crop is sowed. Each drip emitter is equally spaced 20 cm (cm) apart. The online dripper line's building design is depicted in Fig. 1. The soil moisture sensors are inserted into the soil up to 5 cm deep at the sowing stage, 17.5 cm deep at the second stage, and 28.5 cm deep at the third stage of development of the tomato plant [49,50].

In conformity with some existing research findings [51], for measuring the total volume of water usage for the tomato crop at different stages, the EVT_C value has been measured using the real time data. "Penman-Monteith" is an effective method for determining crop water demands [52]. It integrates meteorological data such as sun radiation, wind speed, humidity, and temperature. It determines how much water a plant needs, prevents over- or under-irrigation, reduces water usage, and maximizes crop production. The Soil moisture-based "Penman-Monteith" (SMPME) is divided into two parts. First method is crop reference evapotranspiration (RET_O) and crop coefficient (KR_C) calculation, and the second method is EVT_C and soil moisture. RET_O is calculated based on FMU climate information specific to a region where tomatoes are planted. This climate data includes temperature, humidity, wind speed, solar radiation, elevation, longitude and longitude. The KR_C is calculated based on the field monitoring unit (FMU) region soil moisture,

Table 3
Tomato crop coefficient values of different growth stages.

KR_C initial	KR_C growing	KR_C mid	KR_C last
04 to 0.5 10 to 30 days	0.7 to 0.8 20 to 40 days	1.05 to 1.25 30 to 40 days	0.8 to 0.9 25 to 30 days

FMU capacity, crop type and crop growth stages.

$$RET_o(\text{mm}) = \frac{0.408\Delta(Rd_n - SG) + \gamma\left(\frac{900}{\overline{Temp}+273}\right)WS_2(VP_s - VP_a)}{\Delta + \gamma(1 + 0.34WS_2)} \tag{5}$$

In the equations (5), the crop surface's net radiation is designated as Rd_n, SG is the density of the soil heat flux for a day period and both Rd_n and SG units are measured in megajoule per square meter and per day ($MJ/m^{-2}/day^{-1}$). $\overline{Temp}$ is the average temperature, WS_2 is the wind speed in meter per second (ms^{-1}) at 2 metres foremost of the water's surface for a given day. VP_s is the saturation vapor pressure, and VP_a is the actual vapor pressure which creates the absence of vapor pressure (VP_s - VP_a) and it is measured in kPa. Δ is the axial orientation of the vapor pressure curve and y is the constant value for psychometric analysis measured in (kPa $^{\circ}C^{-1}$). The KR_C value for the tomato crops at different stages has been measured using the Eq. (6) [53]. The general reference value of KR_C at different stages for the tomato crop has been acquired from FAO irrigation and drainage data [54] listed in Table 3. Depending on the soil moisture, temperature, wind speed, and solar radiation for a specific region, these values can be changed at each stage of growth.

$$KR_C = \frac{EVT_C}{RET_O} \tag{6}$$

Eq. (6) shows that EVT_C can be calculated as a measure of the accurate water used for a tomato crop during a particular stage of growth and RET_O is the reference evapotranspiration which is calculated from Eqs. (5). The accurate rate of water used as EVT_C for calculating KR_C value is being validated using two approaches i.e. monitoring the soil moisture sensor value initial and after drainage out and drainage rate value. A 4.7-inch vertical hole has been drilled at the particular FF area under the soil. A drainage container is installed at the bottom of the soil container through which the water from the soil container will drain out. The container of soil in which the hole was filled with water is monitored every 15 min until the entire hole is out of water. A drainage rate is computed from the level of water in the drainage container (inches), the total time required to drain the drainage container (hours), and the soil moisture level at a given time. In the second method, the starting and finishing values of each KR_C growth stage are obtained using the current day value (the day when tomatoes are seeded). Every expansion will have the same result from this strategy. After obtaining the RET_O and KR_C, the EVT_C is determined using Eqs. (7) as the total amount of water needed for the specific crop.

$$EVT_C = KR_C * RET_O \tag{7}$$

The algorithm 2 water_requirement() function assesses the volume of water a tomato crop demands at every growth stage. The entire process is repeated inside a while loop to compute KR_C values based on the starting days when tomato crops are sown or planted. An EVT_C is computed from KR_C and RET_O values measured at a certain growth stage to calculate the total water requirement for irrigating the tomato crop for the day. The Optimal Soil Wetness Closeness Policy (OSWCP) objective function is generated using the PPO on-policy Reinforcement learning (RL) approach, which increases an anticipated advantage while maximizing a surrogate objective. The RL "Q-Network" has been applied in the existing irrigation system for controlling sprinkler irrigation [55]. But in this, the proposed algorithm has been extended as OSWCP using RL "PPO" to optimize the control of underground drip irrigation each day during time interval, estimate various stage climate condition, measure the soil moisture, N,P, and K level. Using RL PPO to optimize drip irrigation systems has a benefit of not requiring the inclusion of any pre-existing rules or hypotheses about soil water status. Instead, it can learn by analysing data on when irrigation and the soil nutrient level policy are being managed [56]. Soil moisture and soil heat variability can be handled by the irrigation agent (multi-layer perceptron) controller, which is capable of training despite labelled output data. Throughout the day, the irrigation process takes place at specific intervals during the morning (6 to 8AM), afternoon (12 to 1PM), and evening-time (4 to 6PM). To prevent various illnesses, regulate soil moisture and evaporation rates, and maintain a less hot environment for tomato FMU, the irrigation time intervals have been chosen to be in the morning, afternoon, and evening [57,58]. By considering the temperature, humidity, soil moisture, RET_O, EVT_C, and KR_C values of the tomato crop, the OSWCP algorithm decides each day when the drip irrigation should be turned on. To make a decision, the circumstances are as follows:

- Soil moisture, RET_O, EVT_C, KR_C, N, P, and K weight are among the constrained collection of states that compose to P.
- The action vector W contains the calculated total water requirement for irrigating the tomato crop in the FMU during specific time intervals (6AM to 8AM, 12PM to 1PM, and 4PM to 6PM). The difference between the current EVT_C and soil moisture value is used to estimate the total water requirement. The SV actuator is automatically activated to water the tomato plant when the RL agent outputs an irrigation (1) action based on the current state and time interval.
- T: P * W → P represents the time steps function T. As an illustration, existing EVT_C and irrigation activity are used to estimate EVT_C's next-time steps.
- J is the value of the reward that characterises the execution of a control action. It is expressed as P × W → J.

Algorithm 2
Soil moisture-based evapotranspiration algorithm for using penman-monteith equations (SMPME).

Input: RET_O, plant growth stage
Output: EVT_C (mm/day), KR_C
function water_requirement (KR_C, EVT_C, RET_O)
while true
  current_days = starting days for growing of tomato
  if current_day <= initial_days
/* crop coefficients start and end value of initial stage */
    KR_C = KR_C_initial_val + (KR_C_last_val - KR_C_initial_val) * (current_day / initial_days_stage)
    KR_C_stage = "Initial stage"
/* Crop coefficients start and end value of growing stage */
  else if current_day <= initial_days_stage + growing_days_stage
    KR_C = KR_C_initial_val + (KR_C_last_val - KR_C_initial_val) * (current_day / growing_days_stage)
    KR_C_stage = "Growing stage"
/* Crop coefficients start and end value of mid stage */
  else if current_day <= initial_days_stage + growing_days_stage + mid_days_stage
    KR_C = KR_C_initial_val + (KR_C_last_val - KR_C_initial_val) * (current_day / mid_days_stage)
    KR_C_stage = "Mid stage"
/* Crop coefficients start and end value of last stage */
  else
    KR_C = KR_C_last_val
    KR_C_stage = "Last stage"
  end
/* Crop water requirement millimetres (mm)/day */
  EVT_C = KR_C * RET_O
/* All stage crop coefficient for a current day */
  KR_C = EVT_C / RET_O
end
return KR_C, EVT_C, RET_O

Algorithm 3
Optimal soil wetness closeness policy algorithm for irrigation recommendation.

Input: State p, Action w, Reward j, an initialized policy, π_θ.
Output: Prediction of irrigation during morning, afternoon, and evening for a day.
Reward (J^{target}) = [irrigation_needed, current_time]
for k = 0 #episode (E_P):
/* State the status of PPO environment policy network with initial state factor and initialize the variable up to current policy θ */
  State p = Soil and plant water environment
  $\theta_{prev} = \theta$
/* Iterate over the number of time steps within each episode */.
  for t = 0, ..., steps
/* Decides an action depending on the current state p_t by using the current policy parameter θ. The action chosen by the policy is defined by $\hat{w}_t$ */
    $\hat{w}_t = \pi_\theta(p_t)$
/* OSWCP Environment responds to the specified $\hat{w}_t$. It is used to generate the appropriate reward j_{t+1} from the environment and update the state p_{t+1} */
    P_{t+1}, j_{t+1} = envrmt.step($\hat{w}_t$)
  end
*/ The advantage has been calculated as the variation between the value function prediction and the discounted overall future rewards. */
  Calculate $\widehat{W}_t$
  Compute reward j_t = Function_reward_grant (p_t, $\hat{w}_t$, p_{t+1})
  Multiple steps and E_p are performed with the batch size C
/* Adjust the policy parameter so that the OSWCP function for loss is reduced by a gradient descent stage */
  $\theta = \theta$ - w∇ θ IP_{OSWCP} (θ)
end
/* function to compute the reward based on each KR_C stage */
Function_reward_grant(p,w**)**
irrigation_needed = p[EVT_C] - p[soil_moisture]
if w == 1
  if irrigation_needed > 0
    if (6 AM <= current_time <= 8 AM) or (12 PM <= current_time <= 1 PM) or (4 PM <= current_time <= 6 PM)
      Return maximum reward for the particular action w and state p, return p [N, P, K level]
    else
      Return penalty for the wrong action time w, p [N, P, K level]
  else
    Return penalty when irrigation_needed action w is less then 0, p [N, P, K level]
  end
else
  Return reward for the particular action w and state p (for no irrigation)
end

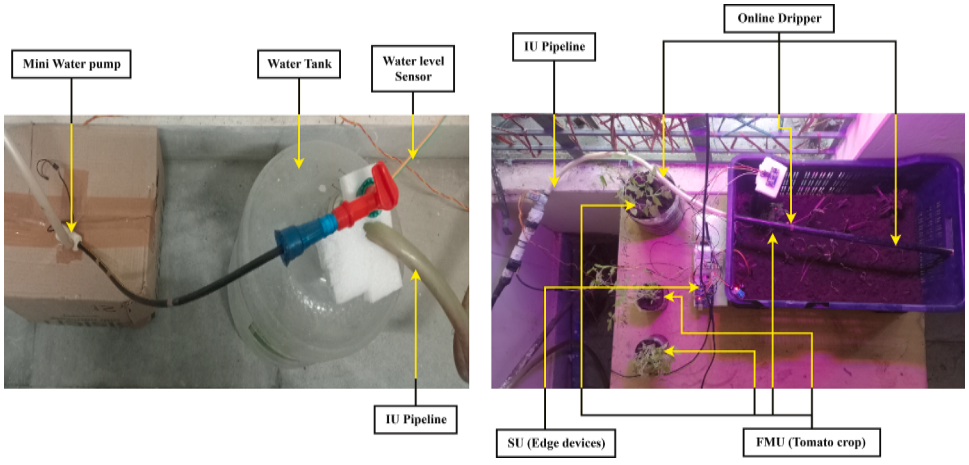


Fig. 4. Tomato crop field monitoring unit prototype setup.

To choose the efficient control policy, the OSWCP model interacts with the environment. In each time step t, the control agent picks an action $W_t = w$ according to policy, considering the present circumstances $P_t = p$. In OSWCP, a reward function has been constructed to convey the goal of attaining maximum tomato plant growth while using the least amount of water. As explained in the introduction subSection 1.1 and Table 1, balancing the temperature, humidity, soil moisture, RET_O, and EVT_C values between the H and B levels is essential to boost the growth rate of tomato plants. This value difference between these two levels has been utilized in Reward J as a stand-in for watering the tomato plant and spurring development. The OSWCP objective function is defined in Eq. (8) for each state P and action space W to minimize the loss function and to optimize crop productivity and irrigation policy.

$$IP_{OSWCP}(\theta) = \hat{D}_t[\min(X_t(\theta)\ \widehat{W}_t,\ clip(w_t(\theta),\ 1-\mathscr{E},\ 1+\mathscr{E})\widehat{W}_t)] \tag{8}$$

The observable anticipation over timesteps is denoted by $\hat{D}_t$ in Eq. (8). The policy variable is θ and the evaluated advantage at time t is symbolized by $\hat{W}_t$. The hyperparameter varying from 0.1 to 0.2 is conveyed by €, and the value according to the upper and lower bound values is the clip. Equations (9) define the $\hat{W}_t$, an advantage function that is utilized to assess the instantaneous advantages of action from a particular state.

$$\widehat{W}_t = \sum_{k=0}^{\infty} \gamma^k j_{t+k} \tag{9}$$

$$w_t(\theta) = \frac{\pi\theta\ (p_t|w_t)}{\pi\theta_{prev}(p_t|w_t)} \tag{10}$$

In Eq. (10), $\pi\theta(p_t|w_t)$ is the updated policy according to the loss function, $\pi\theta_{prev}(p_t|w_t)$ is the previous policy that was used to gather data involving environmental impact. The spread move refers to the distinction between the policy that is updated and the policy that acquires input. The ratio $w_t(\theta)$ uses importance collection to adjust for this drifting. Since the probability of these two ratios can reach high values and cause training to become unstable, the ratio is clipped with an €. Soil moisture, EVT_C, KR_C, RET_O N, P, and K, are all represented by the state vector p. Step orbiting w is the different possibilities an agent might undertake €{1,0}, where 1 signifies set of desirable irrigation requisite during morning 6 – 8AM, afternoon at 12 – 1PM, evening 4 – 6PM with total water volume requirement, and 0 signifies that there is no irrigation recommended. The three scenarios have been taken into consideration in this reward function to reduce the amount of water used without negatively affecting the tomato plant. First, when the soil moisture is low, EVT_C is high, RET_O is high, and KR_C is between growth, and fruit stage. This signals that the tomato plant is experiencing drought stress in the ith drip irrigation region and calls for irrigation. For the scenario of measuring nutrient level it measure N, P and K level is within the required threshold or not. In the second scenario, there is no need for irrigation because the volume of water utilized for irrigation exceeds what the plants require and the water consumption is excessive when the soil moisture is high, EVT_C, and RET_O are low. Thus, in this instance, sickness may have an impact on plants. In the third instance, the tomato plant is strong when soil moisture range between H and B. To conserve water, in this instance, the soil moisture is balanced.

The overall irrigation scheduling is fully automated and utilized to suggest watering each day during morning, afternoon and evening based on the temperature, humidity, wind speed, solar radiation, RET_O, EVT_C, KR_C, soil moisture, N, P, and K value which has been developed using Algorithms 2 and 3 as Optimal Soil Wetness Closeness Policy (OSWCP). Algorithm 3 uses a reward J function to compute the instantaneous reward that the agent gains for accomplishing action W in state P. The irrigation action's ability to sustain soil moisture ultimately distinguishes this reward framework, and ineffective activities result in penalties. When the interval for irrigation is predicted, the water flow meter will begin to calculate the amount of water used to irrigate the tomato crop, and the solenoid irrigation valve will automatically unlock to begin watering them. As soon as the necessary amount of water has been

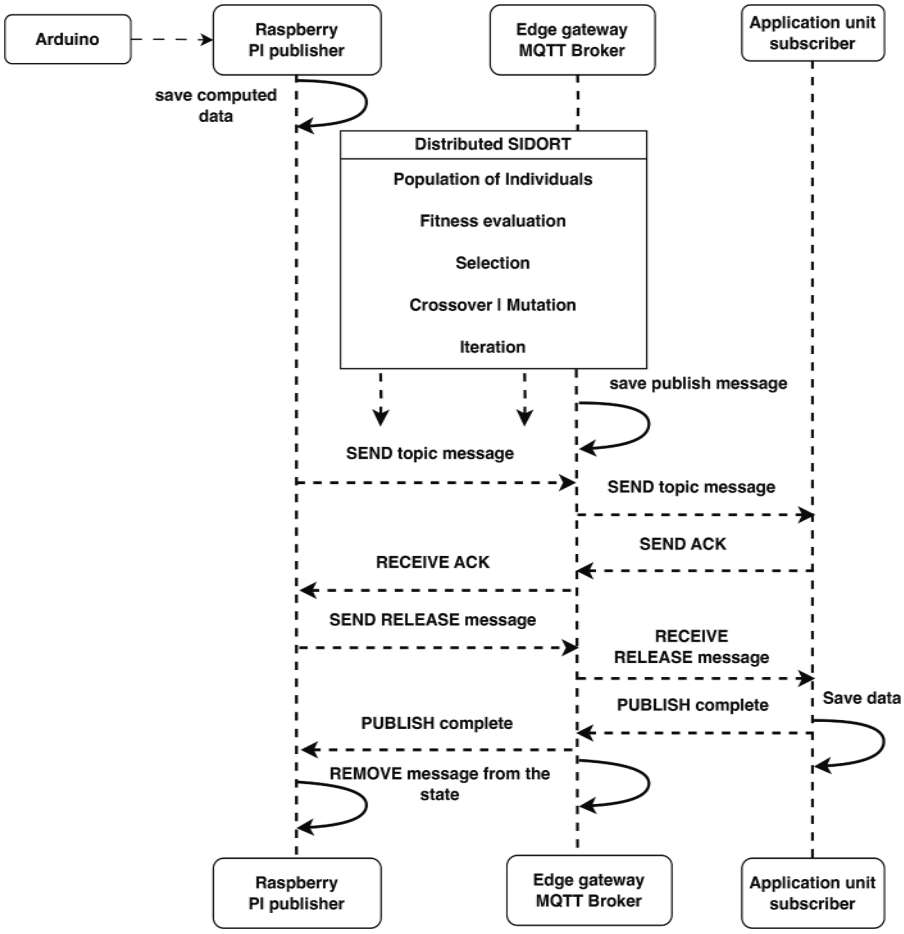


Fig. 5. MQTT data flow architecture.

irrigated and the soil moisture content reaches an ideal level, the solenoid valve will be deactivated to cease irrigation. Eq. (11) automates the measurement of the total volume of water consumed.

$$G_{max} = \sum_{l=1}^{24} (WR_l * WD_l) \tag{11}$$

Eq. (11) states that G_{max} is the sum of the amount of water usage in 24 h (M/L), WR_l is the water rates of flow in the lth hour (measured in M/L per hour), and WD_l is the water flowing duration in the lth hour (measured in hours). Every time an irrigation period occurs at a specific time based on the OSWCP prediction, the water flow meter and SV are turned on to irrigate, measure the water flow rate WR_t, and update the water usage for the current hour (WD_t). Next, by computing the difference between the total tank capacity (TC) and the total water consumed (G_{max}) in a day, the current water level status (WLS) is updated. The water pump fills the tank whenever the WLS value is less than the minimum water level (MIWL) threshold value. If the WLS value exceeds the maximum water level (MAWL) threshold, the water pump stops filling the tank. To determine whether the tomato crop is receiving accurate irrigation and nutrients level during that specific time, the height of the tomato plants is measured using an IR haunt sensor and an ultrasonic distance sensor. The proposed IoT-based tomato crop height measurement technique is built by altering the current system that uses an ultrasonic sensor to compute the height of any item [59–61]. This allows for the calculation of the height of the tomato plant during its growth stage. A haunt sensor is positioned at the base of each tomato plant, and an ultrasonic SR04 sensor is positioned 200 cm above the base to provide an exact height measurement. The θ value will be updated by $\tan^{-1}$ each time the haunt sensor finds the tomato plant base. Dividing the top height by the base height value is equal to $\tan^{-1}$. The overall height of the tomato plant is determined by multiplying the top height through $\tan(\theta)$. The top height is the ultrasonic distance sensor value. In Fig. 4, a prototype setup of the tomato crop FMU has been shown, where the overall automated irrigation, soil nutrient level, water tank monitoring and plant height monitoring system are being tested.

```
pi@raspberrypi:~/Desktop/SDN_irrigation $ python3 tomato_crop_sensor_data.py
LoRa receiver connected..
ARD Node Connected ID: LoRa-001
payload:  [50, 53, 46, 53, 48, 124, 53, 54, 46, 53, 48, 124, 53, 54, 54, 124, 49
, 124, 50, 46, 55, 55, 124, 48, 46, 48, 48, 124, 48, 46, 48, 48, 124, 48, 46, 48
, 48]
payload Size:  37
Last updated:  2023-11-24 08:39:57.352934
LoRa receiver connected..
ARD Node Connected ID: LoRa-001
payload:  [50, 53, 46, 53, 48, 124, 53, 54, 46, 57, 48, 124, 53, 54, 53, 124, 49
, 124, 50, 46, 55, 54, 124, 48, 46, 48, 48, 124, 48, 46, 48, 48, 124, 48, 46, 48
, 48]
payload Size:  37
Last updated:  2023-11-24 08:40:17.504532
LoRa receiver connected..
ARD Node Connected ID: LoRa-001
payload:  [50, 53, 46, 53, 48, 124, 53, 54, 46, 56, 48, 124, 53, 54, 56, 124, 49
, 124, 50, 46, 55, 56, 124, 48, 46, 48, 48, 124, 48, 46, 48, 48, 124, 50, 56, 46
, 52, 53]
payload Size:  38
Last updated:  2023-11-24 08:40:37.638400
LoRa receiver connected..
ARD Node Connected ID: LoRa-001
```

Fig. 6. Connected LoRa based edge devices with Lora based edge server.

3.3. Optimum data transmission between edge and cloud

The straightforward transaction, integration, and analysis of all FMU data is made easier by the Smart irrigation data optimization and robust transmission (SIDORT) Message Queuing Telemetry Transport (MQTT) method. In this study, we adopt a Distributed evolutionary computation framework (DEAP) method based on Genetic algorithm (GA) [62] to design a SIDORT MQTT framework for optimizing the smart irrigation data transmission between distributed edge gateway and cloud server. It evaluates the edge-to-cloud data transmission process by fitness function and provides a feedback mechanism to recognize lost or received data. The Fig. 5 shows the SIDORT MQTT data transmission architecture diagram. SIDORT MQTT expedites and optimizes communication at every layer of the technology stack. A distributed embedded LPWAN [63] based edge server is deployed near each edge device, the edge device sends the FMU data using a sensor and actuator. First an machine to machine (M2M) communication is established to gather FMU sensor data, then the ECU processes and computes the sensor data and saves it in the local storage (MongoDB). As soon as the client subscribes to a "during harvesting" topic, such as irrigation scheduling alerts, farming field climate condition, soil moisture and nutrient levels, water levels in the water tank, total water consumption, water requirement for a day and tomato yield status, the Raspberry Pi as edge server (ECU) processes these results, and then publishes it to the MQTT broker. The smart irrigation application real time data are transmitted in the format of "JavaScript object notation" (JSON) object array [64]. Smart irrigation data optimization and robust transmission (SIDORT) aims to optimize data transmission by optimizing resource allocation, efficiency, and adaptability to varying network conditions. MQTT data transmission is scheduled based on GA computation.

1. First, parameters (genes) related to data transmission priorities, sizes, and frequencies are set for each individual (chromosome) in the population.
2. Every chromosome in the population is a feasible data transmission solution, and each one is unpredictably established.
3. Performance metrics such as latency, throughput, total message sent, reliability, memory allocation, and CPU consumption are used to evaluate the individual based on the fitness function.
4. The chances of being selected for offspring are higher for chromosomes with the best fitness results.
5. As a result of pairing and merging the parameters that produce offspring, a fresh setup and biological crossover get processed for the chosen chromosomes.
6. A mutation operator is used to flip the individuals randomly and explore new configurations while preserving genetic variation.
7. In each generation, steps 3 – 6 take place and as a population expands, the average level of fitness increases.
8. The best configurations for smart irrigation application data transmission from edge to cloud are selected after completing a specified number of generations. ThingsBoards dashboards as ALU can be accessed from any location through mobile or laptop to monitor the FMU information published by the ECU layer.

4. Experimental results and analysis

A smart irrigation system testbed based on edge-based computing has been simulated using Arduino C++ and Python. The entire system's data are stored in a "MongoDB" database for both local and cloud access. Smart irrigation data optimization and robust transmission (SIDORT) MQTT has been utilized for real time data transmission from ECU to ALU. The Thingsboard dashboard has been used to demonstrate the overall system results.

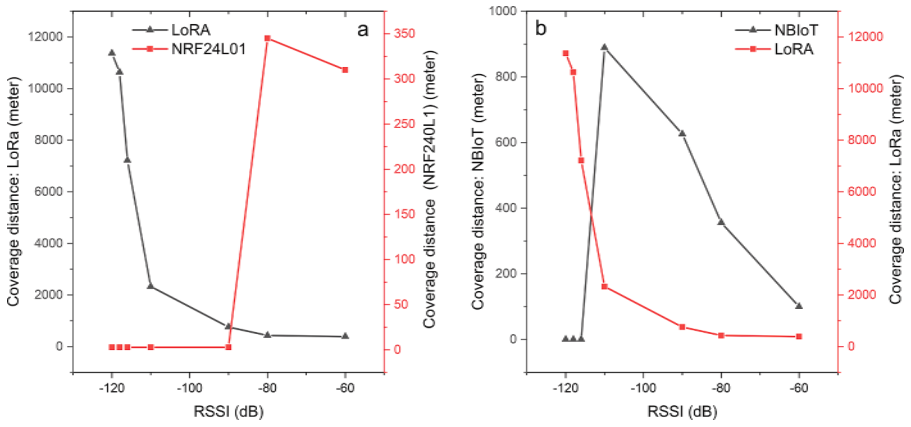


Fig. 7. Coverage distance of connected edge devices with LoRa, NBIoT and NRF24L01.

Table 4
Comparison of different communication technologies in LPWAN sensor network.

Network module	RSSI (dB)	Maximum transmission rate	Coverage distance (meter)
LoRaWAN (LoRA)	−140 to −60	Up to 450 Kbps	More than 12,000
NBIoT	−100 to −60	250 Kbps	Maximum 2000
ZigBee	−100 to −80	230 Kbps	Maximum 100
NRF240L1	−90 to −70	850 Kbps	Maximum 105
ESP32 WIFI	−90 to −50	920 Mbps	Maximum 80
Bluetooth	−90 to −60	5 Mbps	Maximum 100

4.1. Point to point connection validation

The Linked least traversal (LLT) algorithm discussed in the subSection 3.1 was utilized to establish the distributed edge IoT sensor network. The connection between the Arduino and Raspberry pi (RPI) nodes is established via the "LoRa sx1278 ra-02″ 433 MHz transmitter device and to achieve successful transmission in real-time, the LoRa radio frequency library has been used for handling error-free data transmission. Fig. 6 shows that there is an Arduino LoRa-001 edge device coupled to an LoRa-001 RPI node. The payload is a series of bytes that includes commands, sensor reading, and Arduino device information. The last updated is the date and time at which the Arduino node LoRa-001 is connected with the RPI node. The connected Arduino edge nodes serving as SU with RPI operating as edge servers will continuously transmit sensor data.

Fig. 7 shows the comparative analysis in graphs (a) and (b) to validate the long-range transmission distance performance of LoRa, “NRF240L1” and narrowband IoT (NBIoT) module based on RSSI in decibels (dB) value. Fig. 7(a) illustrates that the LoRa module covers distance more than 10,000 m at −80 dB RSSI, while the NRF240L1 module covers only about 300 m using the LLT algorithm for establishing a connection between Arduino and RPI. Based on Fig. 7(b) analysis, the NBIoT module has a range of less than 1000 m, compared to the LoRa module at −100 dB RSSI. The connected Arduino edge nodes serving as SU with RPI operating as edge servers will continuously transmit sensor data. The LoRaWAN module is used to establish this distributed machine to machine (M2M) transmission using the LLT method to transmit FMU data to multiple edge IoT gateways in real-time, provide maximum long-range link coverage, reduce latency and data loss, and enable efficient communication without the need for an internet connection. This M2M communication method will be more efficient as Arduino and RPI devices increase because the payload size will be smaller, resulting in less data loss and latency. Table 4 shows the comparative results of the used low power wide area network (LPWAN) sensor network with another network on various parameters.

A comparison of distributed LoRaWAN-based M2M communication with other network modules such as NB-IoT, Bluetooth, and ESP32 WIFI modules shows that distributed LoRaWAN-based M2M communication has better power consumption, latency, coverage range, and stable connectivity.

4.2. Automated underground drip irrigation performance evaluation

The received sensor data incorporating real-time API data of particular region’s windspeed, solar radiation, latitude, longitude, and elevation values are pre-processed and filtered in the ECU with the current date and time. Automatic underground drip irrigation has been scheduled via a hybrid approach as discussed in subSection 3.2. First EVT_C and KR_C are calculated by applying the soil moisture-based “Penman-Monteith” (SMPME) algorithm and then using the Optimal Soil Wetness Closeness Policy (OSWCP) algorithm irrigation has been scheduled each day at specific interval. A real-time sensor data collection system captures temperature, humidity, wind speed, solar radiation, elevation, longitude, and latitude of the region to calculate the first RET_O value using Eq. (5). Through



Fig. 8. Crop growth from initial to development stage between 1 and 50 days.

algorithm 2, the total amount of water required for a specific region of tomato crops field has been determined based on the KR_C, EVT_C, and real-time soil moisture readings. After obtaining the KR_C and EVT_C values, each day irrigation decisions and N, P, and K level were made through the OSWCP approach with current Soil moisture, EVT_C, KR_C, N, P, and K. Fig. 8 displays the growth status for the tomato crops at their beginning and development stage between 1 and 56 days. The proposed irrigation scheduling algorithm experiment was conducted from October 4, 2023, to December 30, 2023. Table 5 displays the real time data collected between 1 and 56 days from the edge devices for the region FMU whose LAT value was 24.679182, LOT value was 92.751441, and elevation value was 67.

Table 5 displays the data format that emerged in real-time from an edge node. In which temperature, humidity, soil moisture, KR_C, RET_O, EVT_C, N, P, and K value are the initial state variables. The statistical interpretation of sensor data is presented in Table 6 after pre-processing.

Optimal Soil Wetness Closeness Policy (OSWCP) Reinforcement learning (RL) has been trained for irrigation recommendations. The parameters used during training the model are policy = "MlpPolicy", total_timesteps=10,000, learning_rate = 0.0003, dataset shape = (992, 9), and clip_range = 0.2. After training, the model has been saved by passing the model object to Pickle's dump() function. The saved pickle objects files were loaded into the edge server so that the edge server as Raspberry Pi can intelligently make irrigation decisions in real time with the most recent real-time data to anticipate irrigation each day at particular time during morning, afternoon and evening. The OSWCP RL model's effectiveness in recommending irrigation has been validated with the RL Q-neural network (Q-NN) model in the Fig. 9 graph based on accumulated reward and error at each training step. Q-NN and OSWCP utilize deep reinforcement learning (RL) methods to improve irrigation as discussed in subSection 3.2. A Q-NN estimates an action based on the value of the current state and has minimal robustness and stability. In contrast, the OSWCP RL model optimizes policy immediately regardless of data size.

Fig. 9 shows that the OSWCP model performs better than the Q-NN RL model. The OSWCP model had an average accumulated reward of 33.67 and an error score of 0.8867, while the Q-NN model had an accumulated reward of 19.26 and an error score of 4.26. The accuracy of controlling underground drip irrigation averaged 98.48 % during training, 97.88 % during validation, and 98.18 % during testing. The graph demonstrated in Fig. 10 shows the variations of EVT_C (estimated water requirement) soil moisture, temperature, humidity, wind speed, and solar radiation for each day during specific time interval in each growth stage (KR_C) of the tomato crops.

In the Fig. 10 graph, EVT_C (b), soil moisture (SM) (c), temperature (a), humidity (d), KR_C (g), wind speed (f), and solar radiation (e) were observed at 11:30:08AM as 531.05 mm/day, 56 %, 26.6°C, 87.4 %, 0.745, 2.092 ms^{-1}, and 622 W per square meter (wm^{-2}) respectively. It suggests that tomato crops should get irrigation during afternoon and soil nutrient level as nitrogen (N) 101 represents a low soil nutrient level, phosphorus (P) 85 represents an optimal value, and potassium (K) 75 represents a low value in milligram kilogram (mg/kg) As mentioned in Algorithm 3, the Optimal soil wetness closeness policy (OSWCP) suggests irrigation if the tomato crops demand watering during specific time interval of the day. In Table 7, the OSWCP model prediction performance has been represented at different time intervals as episodes with accumulated reward and mean absolute error (MAE) score of predicted action based on the current optimized input state.

Fig. 10 and Table 7 show the efficiency of the proposed automated OSWCP RL algorithm for scheduling underground drip irrigation for tomato crops. The outcomes show that rewards sometimes increase or decrease. This variation of rewards relies on the changes in the current state environment and action made by the RL agent. The RL model is continuously in the learning process. Optimizing current state temperatures, humidity, soil moisture, EVT_C, RET_O, N, P, and K will result in greater rewards. However, if the soil moisture level is below the minimum threshold, EVT_C and RET_O continue to grow, which results in decreased rewards. In this case, the RL model recommends for irrigation. To validate the results obtained from the proposed OSWCP automated underground drip irrigation and plant growth monitoring system for tomato crops, manual irrigation methods were also used to monitor soil health and plant growth after or before irrigation during the day in the morning and evening. In Fig. 11, the graph analysis determines the soil moisture status and the height of tomato plants based on the day and time of OSWCP and manual irrigation. The soil moisture validation graph (a) in Fig. 11 shows that manual irrigation does not achieve a balanced soil moisture level. In the morning and evening intervals of a particular day, soil moisture levels of the manual irrigation method are moving toward low or high levels. In

Table 5
Real time data collection in edge node between 1 and 56 days.

Days	Last updated DD-MM-YYY HH: MM: SS	Temperature [°C]	Humidity [%]	Soil moisture [%]	RET_O [mm]	EVT_C [mm/ day]	KR_C	Wind speed [ms^{-1}]	Solar radiation [wm^{-2}]	N [mg/ kg]	P [mg/ kg]	K [mg/ kg]
0	04–10–2023 2:32:00PM	29.7	82.6	87	319.64	127.85	0.40	1.41	252	94	38	53
1	05–10–2023 6:38:00AM	27.2	80.4	81	242.05	98.43	0.40	1.23	208	98	35	61
5	14–10–2023 7:03:00AM	25.9	88.6	67	137.36	59.524	0.43	1.08	171	117	42	59
17	26–10–2023 12:00:00PM	27.6	78.8	48	485.06	285.77	0.45	1.44	732	117	43	61
28	31–10–2023 10:48:00AM	26.8	83.2	79	500.62	362.11	0.46	1.07	792	106	39	55
49	27–11–2023 4:28:00PM	24.8	81.8	87	12.87	9.62	0.74	1.4	20	118	31	68

Table 6
Statistical summary of sensor data.

Statistics	Temperature [°C]	Humidity [%]	Soil moisture [%]	RET_O [mm]	EVT_C [mm/day]	KR_C	N [mg/kg]	P [mg/kg]	K [mg/kg]
Mean	28.83	85.15	73.34	88.37	56.31	0.68	111.41	38.80	53.53
Maximum	31.90	98.20	90.00	697.56	668.23	1.81	124	93	146
Minimum	24.12	61.34	48.00	3.26	1.25	0.41	54	24	50
Standard deviations	0.65	3.92	1.34	128.93	51.38	0.11	10.63	3.31	4.48

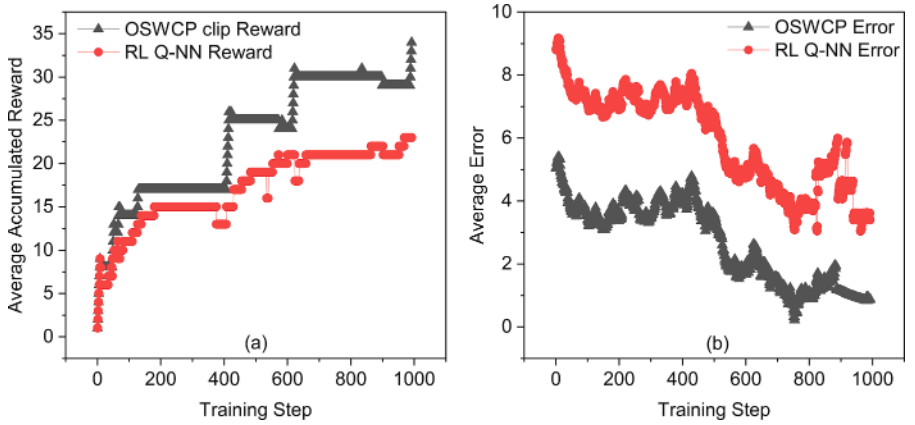


Fig. 9. The average accumulated reward and error by OSWCP model.

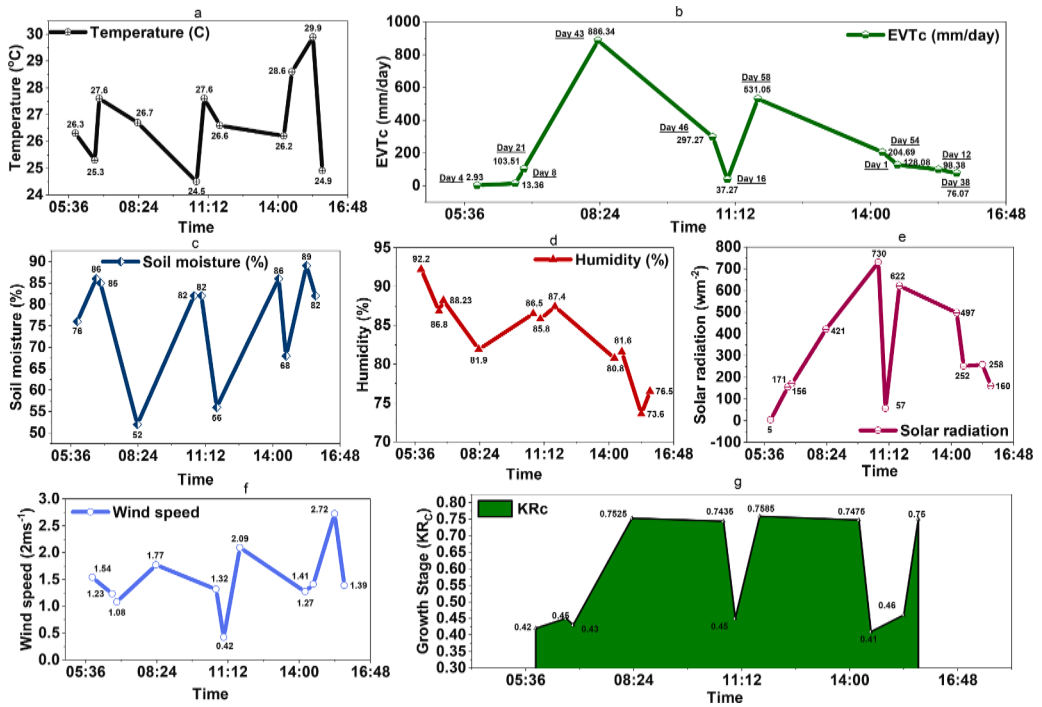


Fig. 10. Time wise irrigation value changes for each growth stage based on the state of input and action performed according to the OSWCP method.

Table 7
OSWCP MAE score and reward for the predicted action based on the current input state transition.

Time interval DD-MM-YYY HH:MM:SS	Action (Irrigation)	Reward	MAE	Soil moisture [%]	Temperature [°C]	Humidity [%]	EVT_C [mm/day]	N [mg/kg]	P [mg/kg]	K [mg/kg]	Tomato plant growth [%]
15–10–2023 07:23:05AM	1	26.12	0.22	58	25.67	74.23	389.45	107	73	114	35%
15–10–2023 11:18:25AM	0	32.01	1.23	65	27.23	79.35	6.23	108	76	120	35%
15–10–2023 02:45:08PM	0	34.14	0.56	65	28.19	79.84	11.97	111	78	119	33%
15–10–2023 05:26:01PM	1	33.67	1.33	62	26.12	81.23	378.67	117	78	119	33%
15–10–2023 09:28:41PM	0	36.13	0.87	76	24.76	81.03	3.34	116	78	118	34%
17–10–2023 06:58:13AM	0	33.24	0.57	65	24.25	78.67	14.45	108	59	98	35%
17–10–2023 12:35:03PM	1	35.08	0.36	59	27.12	79.13	358.25	107	59	98	35%
17–10–2023 05:13:45PM	1	35.23	0.28	60	28.29	79.67	137.78	108	67	111	35%
17–10–2023 11:05:13PM	0	38.08	1.11	75	25.23	81.16	2.98	111	43	112	36%
21–10–2023 12:15:23AM	0	41.19	0.47	72	27.32	78.17	12.67	115	93	134	39%
21–10–2023 05:26:01PM	1	42.26	0.12	63	26.31	80.12	156.45	124	89	131	39%
26–10–2023 07:16:10AM	0	39.27	1.03	78	26.35	76.44	8.09	126	65	84	41%
26–10–2023 05:08:19PM	0	26.67	3.78	57	26.89	83.01	289.68	126	64	81	41%
01–11–2023 09:34:00PM	0	30.12	1.16	69	24.33	78.13	10.12	115	83	121	52%
09–11–2023 06:48:20AM	0	42.24	0.19	82	25.48	80.03	2.01	118	76	126	58%
17–11–2023 07:47:00AM	0	37.12	0.14	79	23.14	81.97	14.23	120	84	117	62%
21–12–2023 12:39:00PM	1	40.19	0.08	64	22.04	80.44	176.08	117	79	94	66%

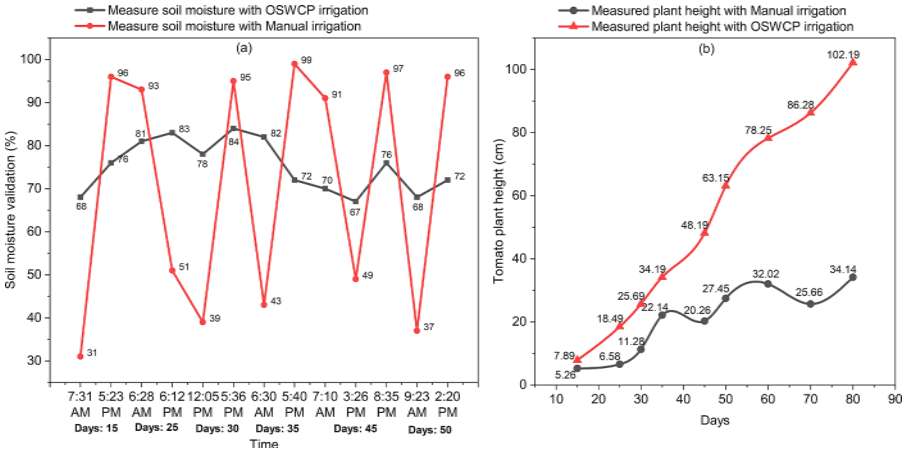


Fig. 11. Comparison of soil moisture and plant height through OSWCP and manual irrigation system.

contrast, in the OSWCP irrigation method, soil moisture level values remain stable between 60 and 85 % on a particular day, which indicates an improvement in tomato plant growth, as shown in Fig. 11 graph (b) with red line curve of OSWCP irrigation tomato plants height is increasing frequently as compared to tomato plants height with manual irrigation.

Soil health status day-wise for both automated underground drip irrigation and manual irrigation methods is illustrated in Fig. 12 graph.

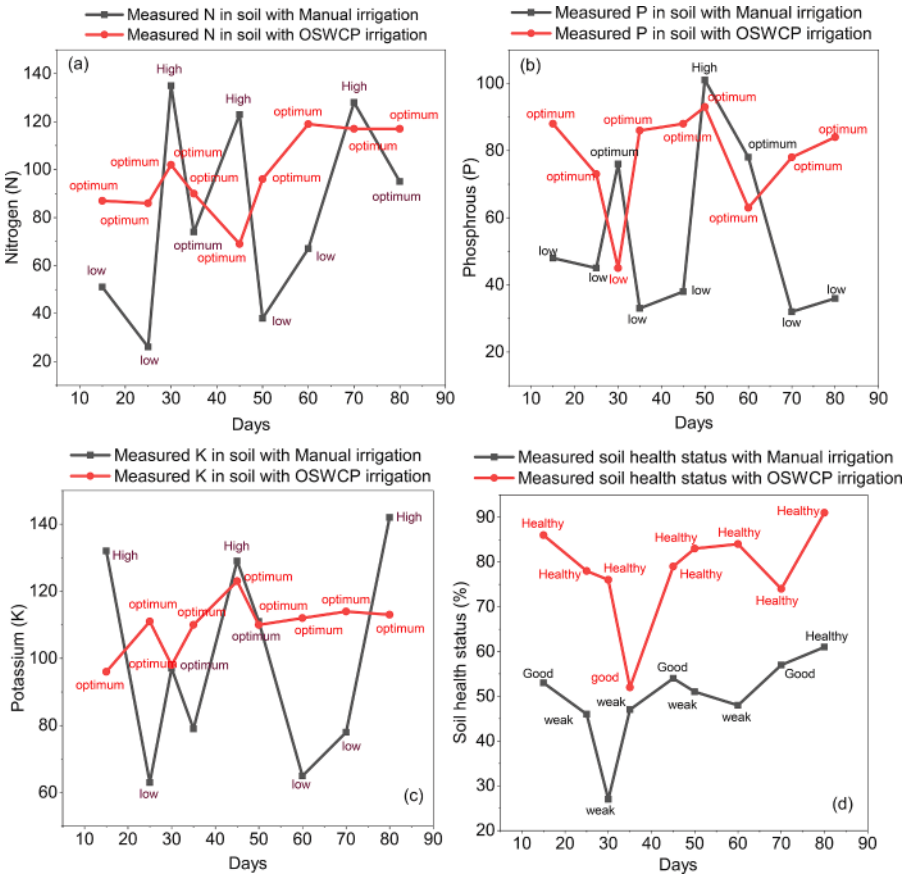


Fig. 12. Comparison of day-wise N, P, K and overall soil health status with Edge IoT-based underground drip irrigation over manual irrigation.

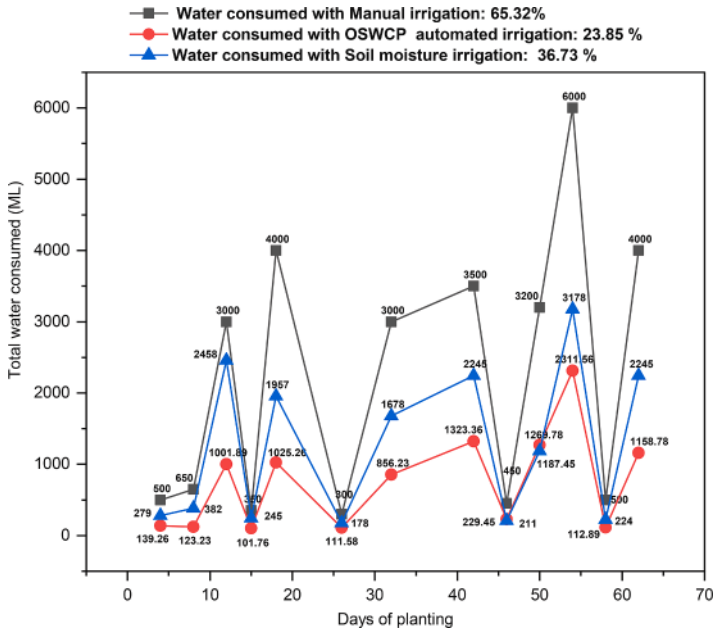


Fig. 13. Comparison of Average water consumption for manual, soil moisture based and proposed automated drip irrigation system.

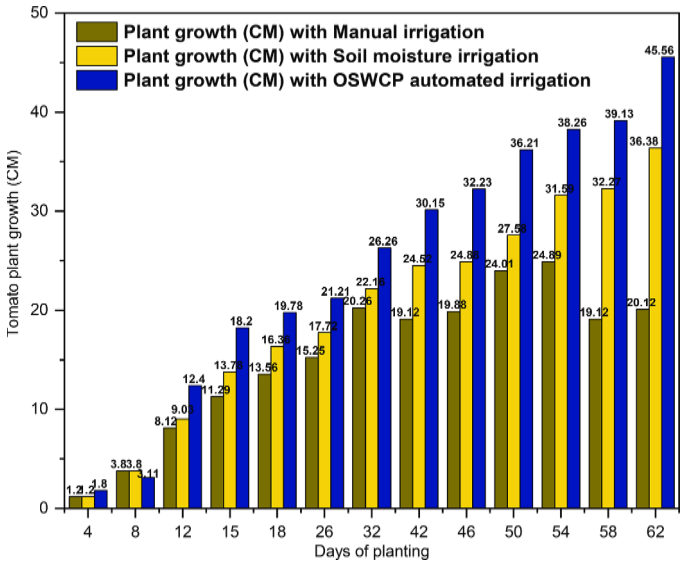


Fig. 14. Plant growth status comparison between automated, soil moisture based and manual irrigation method.

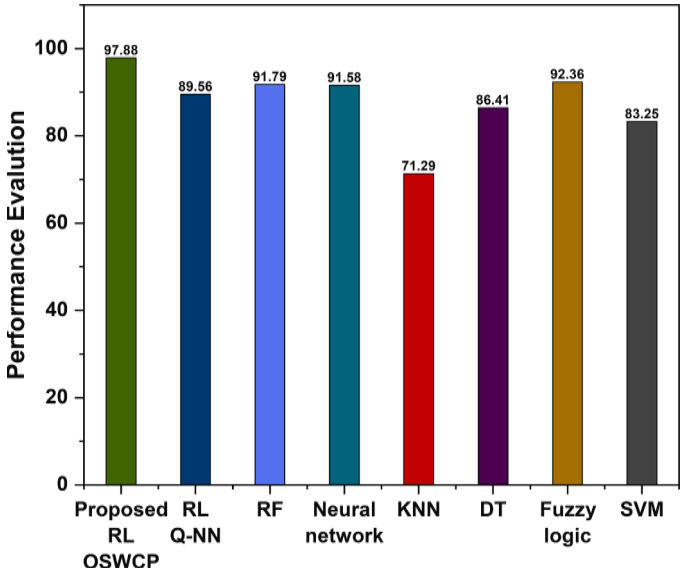


Fig. 15. Proposed RL OSWCP performance comparison with another ML algorithm.

The estimated soil health for tomato crops is visualized in Fig. 12, where graphs (a, b, and c) show the nitrogen (N), phosphorus (P), and potassium (K) levels are often at optimum levels on a particular day of Optimal soil wetness closeness policy (OSWCP) irrigation method from which soil health status is either healthy or satisfactory, as shown in Fig. 12 graph (d) analysis. In contrast, when irrigation is performed manually, soil health improvement is minimum between 10 and 90 days, due to an imbalance of soil moisture and improper or excessive irrigation. The tomato crop is susceptible to diseases when the soil moisture and nutrients are imbalanced [65]. Therefore, the OSWCP automated underground drip irrigation method performs better than the manual irrigation method for balancing soil moisture and soil nutrients. The total volume of water used for the suggested automated underground drip irrigation system, soil moisture based irrigation, and manual irrigation on a specific day is depicted as a graph in Fig. 13, Fig. 14 displays the tomato plant's growth state after manual, soil moisture based and automated underground drip irrigation method are applied. The water utilization of the suggested automated irrigation system has been verified by storing the water in three separate tanks for irrigation. Wherein Eq. (11) has been used to calculate the total amount of water used for soil moisture based, automated underground drip and manual irrigation systems. Fig. 13 displays the total water used for automated drip irrigation as a dark red line, the total water used for manual irrigation as a Gray line and total water used for soil moisture-based irrigation as a sky-blue line, day-wise. When compared to manual and soil moisture based irrigation, it is evident that the adaptive EVT_C OSWCP underground drip irrigation system

Table 8
Comparative table for irrigation prediction results of different intelligent methods.

Method	Temperature [°C]	Humidity [%]	EVT_C (mm/day)	Soil moisture [%]	Time [AM/PM]	MAE score	Irrigation status
Proposed RL OSWCP	26.78	81	325.18	62	7:45AM	1.03	1
Q-NN RL	28.16	77	423.17	73	11:37PM	7.78	1
SVM	30.23	91	623.04	52	2:23PM	12.35	0
RF	27.78	93	337.08	62	5:12AM	3.29	1
Neural network	30.25	79	156.48	78	7:45PM	4.96	1
KNN	30.63	88	455.98	43	12:19PM	14.89	0
DT	28.25	73	568.12	64	11:21AM	6.93	1
Fuzzy logic	28.55	85	786.21	54	4:28PM	3.62	0

Table 9
DEAP framework hyperparameter and SIDORT model configuration.

Register	Tools	Parameters
Mutation	mutGaussian	mu=8, sigma=4, indpb=0.2
Selection	selTournament	tournsize=3
Population	population	Pop_size = 50, generation = 30

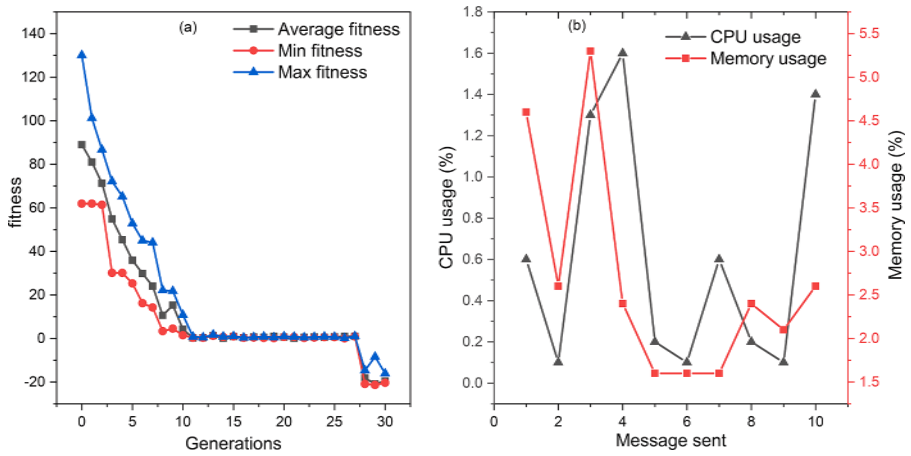


Fig. 16. SIDORT MQTT performance graph analysis.

(shown with the dark red line) uses less water to irrigate tomato crops. Additionally, Fig. 14 demonstrates that, in comparison with manual and soil moisture based irrigation, the tomato plant's growth status improved when the suggested adaptive EVT_C underground drip irrigation system was used for 30 to 60 days.

It is estimated that automated underground drip irrigation systems use 22.85 % of the water, compared to manual and soil moisture based watering systems which use 78.45 % and 41.73 %. According to several investigations previously accomplished [66,67], using IoT-based automated irrigation systems to irrigate crops increases crop yields and revenues while preserving water and improving water quality for the development of tomato crops. Thus, farmers' efforts, resources, and time serve a purpose precisely when utilizing the suggested self-sustaining intelligent edge IoT-based irrigation system, which can be used in both indoor and outdoor farming. The proposed AI-based OSWCP irrigation scheduling algorithm performance has also been validated by comparing it to other algorithms, including Q-NN RL, Random Forest (RF), support vector machine (SVM), Neural Network (NN), and Fuzzy logic. Concerning the recommended AI framework for self-adaptive irrigation systems, Fig. 15 exhibits the performance score of alternative AI models utilizing the same real time sensor dataset and Table 8 shows the irrigation predictions results of different method.

According to Table 8, which compares different methods for predicting irrigation, OSWCP predicts optimal irrigation based on EVTC, soil moisture, and time interval. The performance of the proposed AI OSWCP algorithm is demonstrated in Fig. 15 with histograms in comparison with other AI based methods.

The Fig. 15 demonstrate that the suggested AI model performs better than existing AI based models (Q-NN, SVM, RF, DT, NN, KNN) and fuzzy logic. Additionally, the performance of the recommended underground drip irrigation scheduling performs better when compared to the existing automatic irrigation scheduling research problem [38]. This is because the proposed irrigation scheduling measures tomato crop growth, nutrient and moisture level from the outset and predicts the total need of water based on real time values of temperature, humidity, wind speed, solar radiation, soil moisture, RET_O, KR_C, EVT_C, N, P, and K. The comprehensive edge IoT-based automated irrigation is a customized application for tomato farming fields that includes a weather station, soil health, and

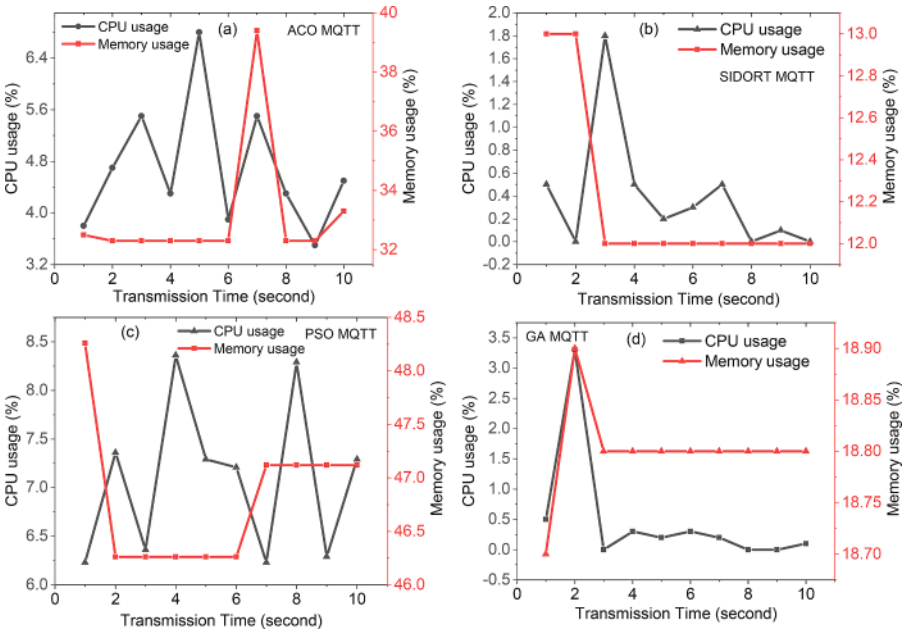


Fig. 17. Comparative analysis of different meta-heuristic approaches for data transmission using MQTT protocol.

Table 10
MQTT Data transmission performance metrics.

Performance metrics	SIDORT MQTT	ACO MQTT	PSO MQTT	Classical GA [62] MQTT
Transmission time (second)	60	85	90	73
Message sent	10	10	10	12
Message received	10	6	5	8
Loss rate (%)	0.02	40.01	50.03	33.33
Latency (second)	6.06	14.16	18.86	9.12
Throughput (Kbps)	0.28	0.07	0.07	0.10
Bandwidth (Kbps)	113.10	111.21	114.01	113.78
Total CPU usage (%)	0.7	3.9	1.09	2.03
Memory Usage (%)	2.78	12.12	8.10	5.23

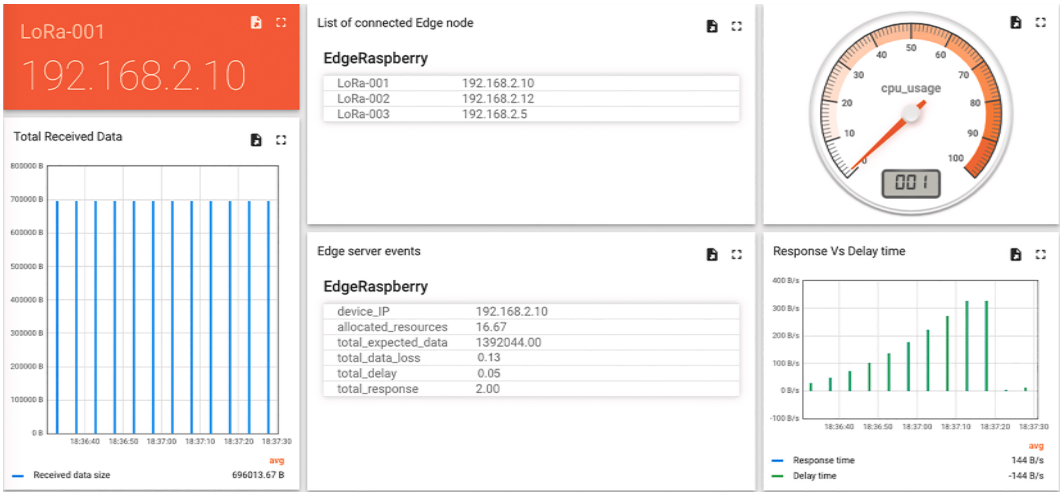


Fig. 18. Connected edge gateway.

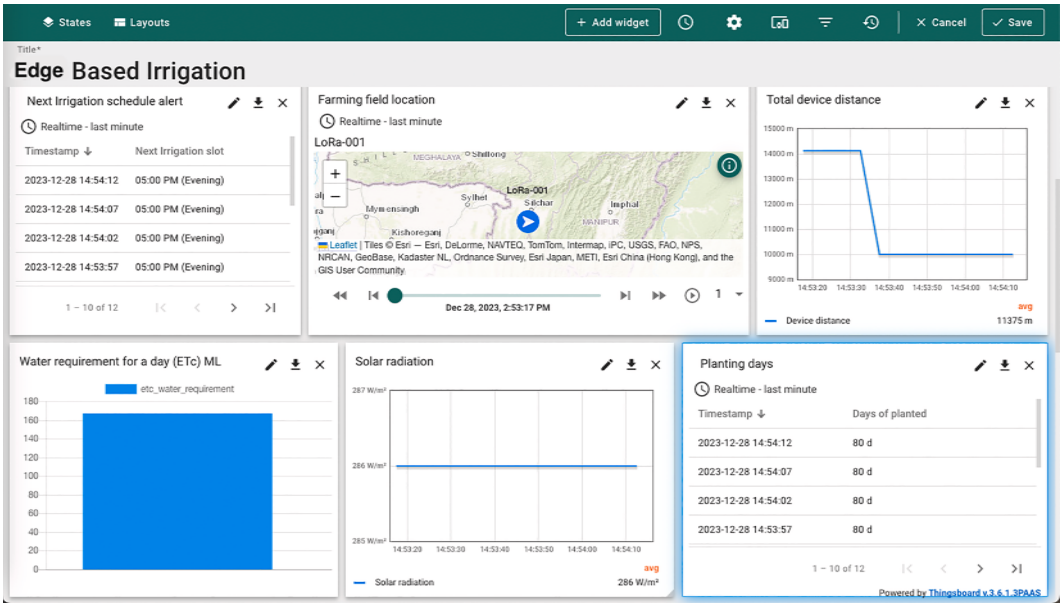


Fig. 19. Notification for scheduled irrigation, current EVT_C, and solar radiation of specific location.

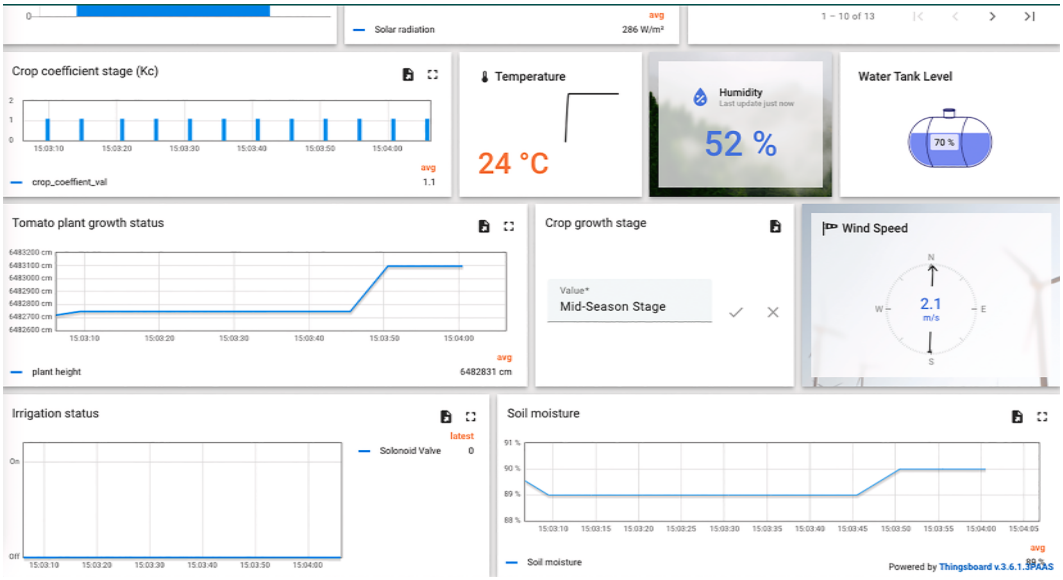


Fig. 20. Current soil moisture, climate, and tomato plant growth of particular farming field.

nutrient level monitoring. It irrigates tomato crops each day at specific time intervals, predicts the total volume of water demand and consumption in a day, measures tomato plant height and monitor water tank level.

4.3. Edge cloud real time data monitoring performance evaluation

Real time data transmission at the Thingsboard cloud server via edge gateway is performed using Smart irrigation data optimization and robust transmission (SIDORT) Message Queuing Telemetry Transport (MQTT) as discussed in the subSection 3.3. Three Raspberry Pi_s used as edge gateway for transmitting real-time data through MQTT protocol. The DEAP python library is used to simulate SIDORT MQTT based on Table 9 hyperparameters.

Fig. 16 depicts the performance of the SIDORT MQTT approach and edge server performance during real-time data transmission in graphs (a) and (b).

The minimum, maximum, and average SIDORT fitness curves for each generation are depicted in Fig. 16(a). Fig. 16(b) shows the

Table 11
Summary of comparative analysis between the proposed smart irrigation system and existing methods.

System	EVT_C combined soil moisture based	Water tank level	Crop growth monitoring	AI based irrigation scheduling	Distributed Edge IoT LoRaWAN standard	Measure Soil moisture & nutrient level	Rechargeable powered energy	Data transmission optimization	Custom system
[27]	No	No	No	Yes	No	Measure only soil moisture	No	No	No
[28]	Yes	No	No	Yes	No	Measure only soil moisture	No	No	No
[31]	No	No	No	Fuzzy logic based with CWSI	No	No	Yes	No	No
[32]	ET_C based only	No	No	No	No	No	Yes	No	No
[36]	Yes	No	No	No	No	Yes, (only soil moisture)	No	No	No
[38]	ET_C based	No	No	Yes, ML (not validated)	No	No	No	No	No
[56]	No	No	No	Yes, (no scheduling)	No	No	No	No	No
Proposed Smart underground drip Irrigation	Yes	Yes	Yes	Yes, OSWCP	Yes	Yes	Yes	Yes, (SIDORT MQTT)	Yes

CPU performance and memory utilization curves for each message transmitted by the edge server. A comparison of the proposed SIDORT MQTT data transmissions results with other optimization methods such as ant colony optimization (ACO), particle swarm optimization (PSO), and classical GA [62] using the MQTT protocol is depicted in Fig. 17 graph and Table 10.

Graphs (a) and (c) of Fig. 17 show that ACO and PSO consume more CPU and memory during MQTT data transmission, whereas SIDORT and GA consume less CPU and memory as shown in Fig. 17(b) and (d). However, SIDORT's performance during MQTT data transmission from edge to cloud server is much better than those of ACO, GA, and PSO, as shown in Table 10 performance comparison.

The ThingsBoard dashboard for overall system monitoring, seen in Fig. 18, Fig. 19, and Fig. 20, serves as a user interface for user-end FMU monitoring. The FMU dashboard in Fig. 19 displays the next irrigation schedule, the total amount of water needed, nutrient level, the level of the water tank, and Fig. 20 shows the current soil moisture, and relevant climate-related information.

4.4. Comparative analysis of the proposed edge IoT-based smart irrigation system with existing systems

A detailed comparison of the proposed edge IoT architecture for scheduling smart irrigation system with the existing algorithm is displayed in Table 11.

According to the experimental results, discussion, and comparative analysis in Table 11 report, the recommended edge IoT-based irrigation system performs better for tomato crops in terms of underground drip irrigation scheduling, total water, and consumption based on climate conditions and soil health. It enables automated irrigation application, long-range edge IoT distributed M2M network, and real-time field monitoring requirement with minimum data loss on latency.

5. Conclusion

This research proposes an edge IoT-based smart underground drip irrigation scheduling model as well as crop growth monitoring for tomato cultivation by addressing the shortcomings of conventional IoT-based irrigation management systems. For establishing communication between the nearest edge sensor device and the edge server of several farming fields in a remote area, a decentralized algorithm known as Linked least traversal (LLT) has been developed. The LLT algorithm will function in all the devices within a span of 11,000 meter. Next, a real-time adaptive AI-based irrigation scheduling algorithm known as Optimal Soil wetness closeness policy (OSWCP) has been developed and therefore increasing tomato plant growth, decrease water waste, and evenly distribute irrigation to each tomato plant. It is used to recommend irrigation every day in the morning, afternoon, and evening and measure soil nutrient levels. An existing AI-based irrigation algorithm has been compare with OSWCP method's effectiveness. It is find that OSWCP has 97.88 % success rate, and it uses 55 % less water as compared to other irrigation method for irrigating the tomato crop. On the DEAP framework, the optimization of FMU data transmission from the edge to the cloud server is simulated using Smart irrigation data optimization and robust transmission (SIDORT) MQTT. Based on the comparison and analysis of SIDORT MQTT results with the existing metaheuristic method, the SIDORT approach achieved more than 0.28 kgbytes per second (Kbps) throughput compared to another transmission method. The proposed edge-IoT-based automated irrigation system is compared and analysed according to the custom smart agriculture application. It overcomes the shortcomings of the existing irrigation algorithm, demonstrating that it is an efficient and low-cost irrigation system for smart agriculture. Regarding further research for the development of self-sustainable automated farming additional features and improvements can be included such as rainfall alert, disease prediction and classification based on nutrient status, venturi systems through underground drip irrigation systems for tomato crops, monitoring of animals, and undesired objects near the tomato plants etc. The advanced software-defined network (SDN) framework can enhance a proposed IoT edge approach by adding data security, device authentication, and LPWAN network routing control.

CRediT authorship contribution statement

Rohit Kumar Kasera: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tapodhir Acharjee:** Writing – review & editing, Visualization, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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